Data Analysis

A Case Study on Stock Market Data

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Oct 24th, 2025

S³Lab
Smart Software System Laboratory


“Prediction is very difficult, especially if it's about the future..”
– Nils Bohr, Nobel laureate in Physics

Data Analysis

Some Concepts

- **What**? - Turning raw data into useful information
- **Why**? - To provide answers to questions being asked by someone

- Even the greatest amount and best quality of data mean nothing if data are not properly analyzed—or analyzed at all.

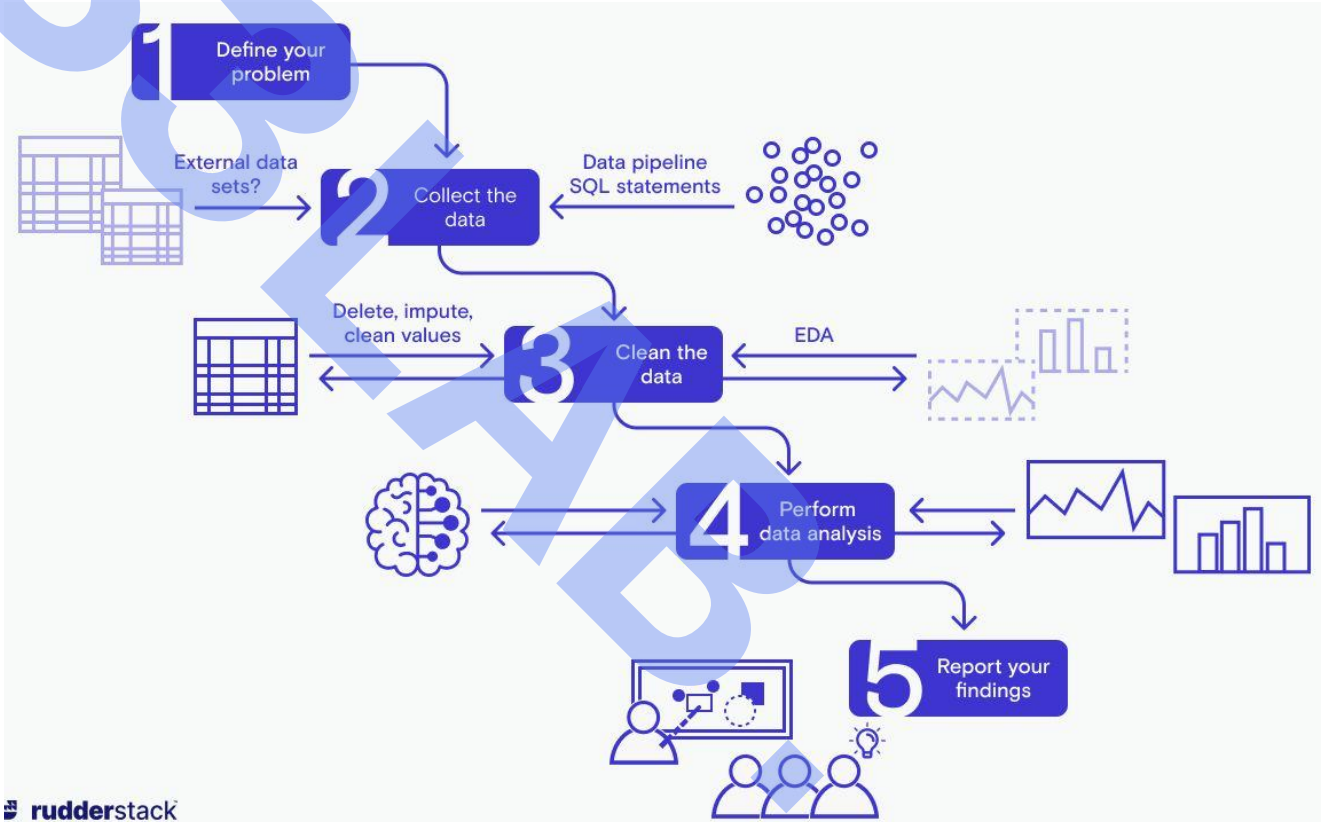
- **Data analytics replaces guesswork with data-driven insights**. It helps you make sense of the past and predict future trends and behaviors, leaving you much better equipped to make smart decisions.

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Data Analysis

Processing Steps

https://www.rudderstack.com/learn/data-analytics/data-analytics-processes/



SOLAB

Data Analysis

Processing Steps - Step 1 : Define your problem

- Define your objective - ‘problem statement’
- Make questions:

  What business problem am I trying to solve?

  Why are we losing customers?

  Which factors are negatively impacting the customer experience?

  How can we boost customer retention while minimizing costs?

- Need soft skills, business knowledge, and lateral thinking

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Data Analysis

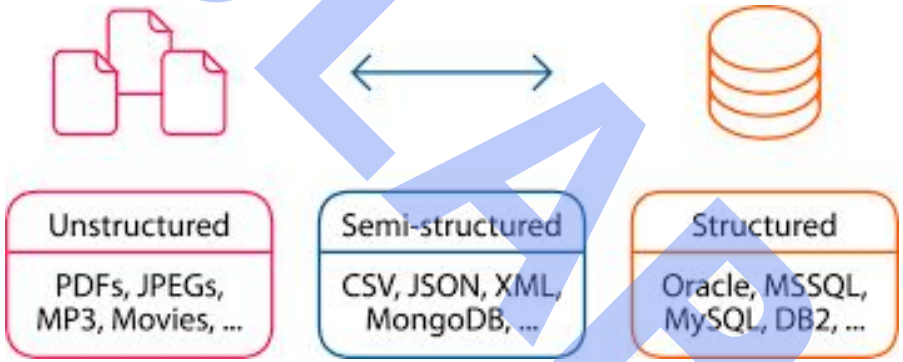
Processing Steps - Step 2 : Collect the Data

- Now you've defined a problem, you need to determine which sources of data will best help you solve it.
- Determining which data you need:
  - This might be **Quantitative** (numeric) data, e.g. sales figures.
  - Or **Qualitative** (descriptive) data, such as customer reviews.
- All data fit into one of three categories
  - **First-party**: from Your Organization, CRM, ERP, or your Owned Database.
  - **Second-party**: the first-party data of other organizations, website, app or social media activity, like online purchase histories, or shipping data.
  - **Third-party**: collected and aggregated from numerous sources by a third-party organization, contains a vast amount of unstructured data points (big data) REF.
  Or the Community like Kaggle, RoboFlow, HuggingFace, …

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Processing Steps - Step 2 : Collect the Data

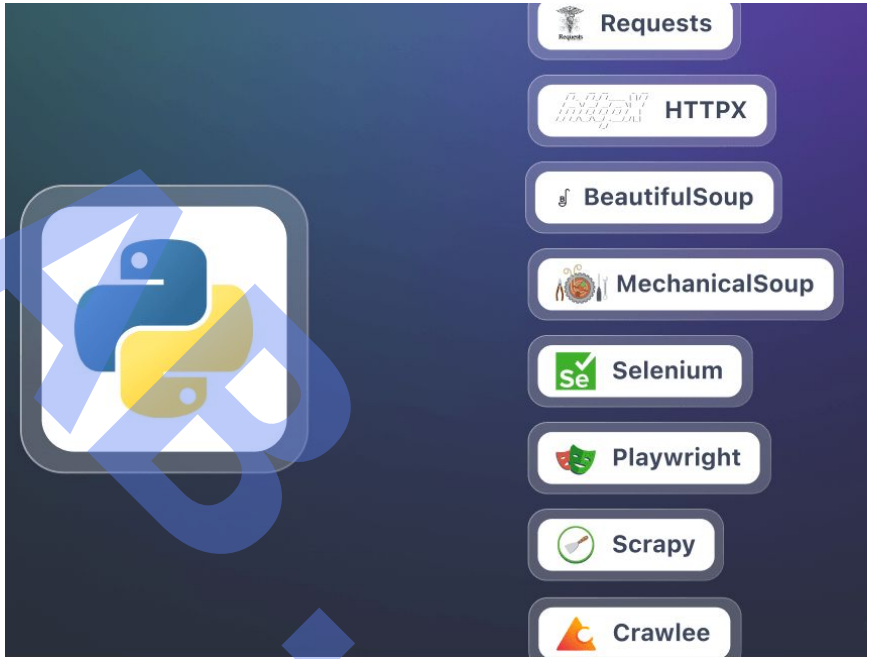


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Processing Steps - Step 2 : Collect the Data

- How to get Data:
  - Web API: Restful API, SOAP, ...
  - Web-crawling and web-scraping
  - Database Connector
  - Download Data as file such as .csv published on internet.



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Processing Steps - Step 3 : Clean the data

- Cleaning the data
  - Removing major **errors**, **duplicates**, and **outliers**
    - 'Iron' (uppercase) and 'iron' (lowercase), 'Iron' and 'Fe'
    - Remove data points that dramatically differ from others in the set
    - Fix contradictory data errors: domain, invalid data (null, wrong data type, …)
  - Removing unwanted data points
  - Bringing structure to your data
  - Filling in major gaps: **remove** the entries, **guess** the missing data, **flag** the data as **missing**
- Take 70-90% of our time

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https://careerfoundry.com/en/blog/data-analytics/what-is-data-cleaning/

Data Analysis

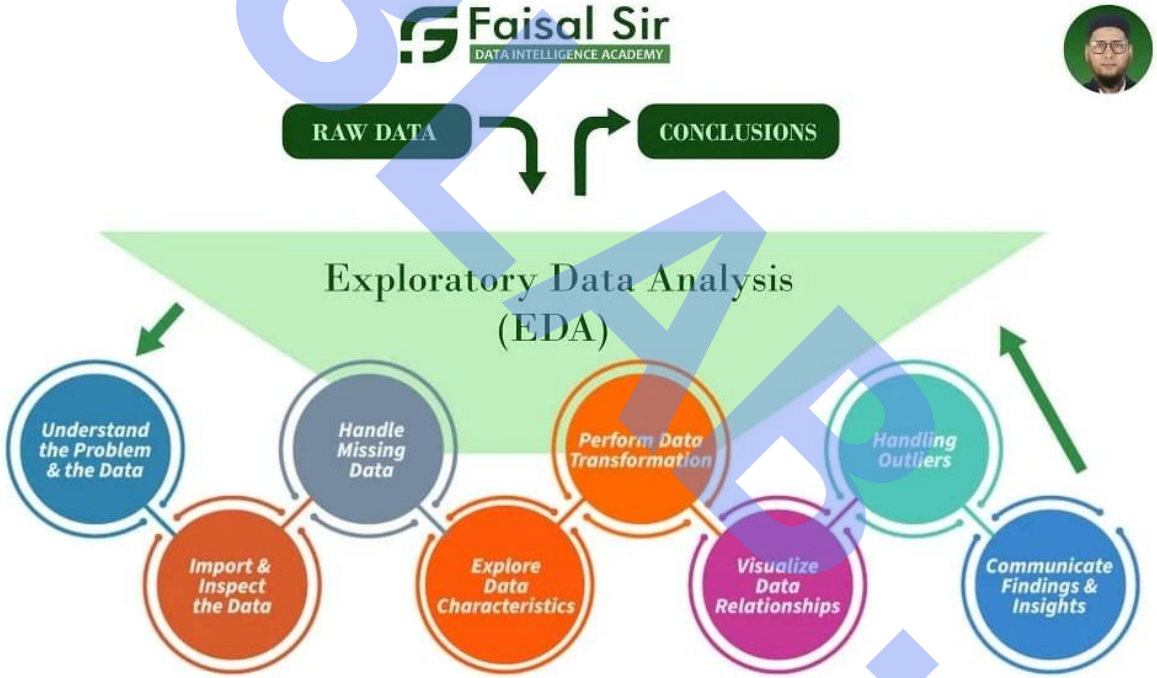
Processing Steps - Step 3 : Clean the data

- Carry out an **Exploratory Data Analysis**
- This helps identify initial trends and characteristics, and can even refine your hypothesis.
- How to do:
  - Python libraries (e.g. Pandas)
  - Enterprise Tool: Data Ladder, Microsoft Excel
  - Open Source Tool: OpenRefine
  - Visualizations

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Data Analysis

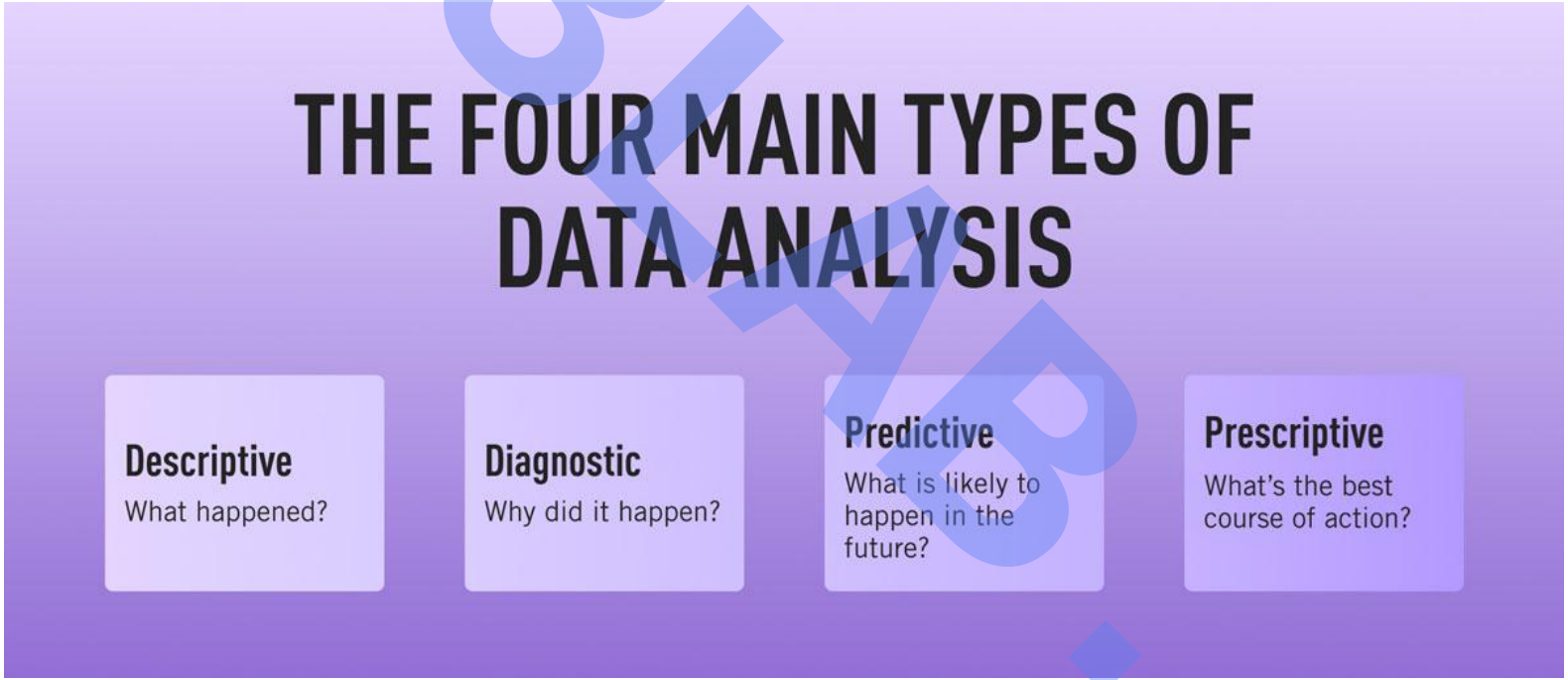
Processing Steps - Step 3 : Clean the data



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Data Analysis

Processing Steps - Step 4 : Perform Data Analysis



https://careerfoundry.com/en/blog/data-analytics/different-types-of-data-analysis/

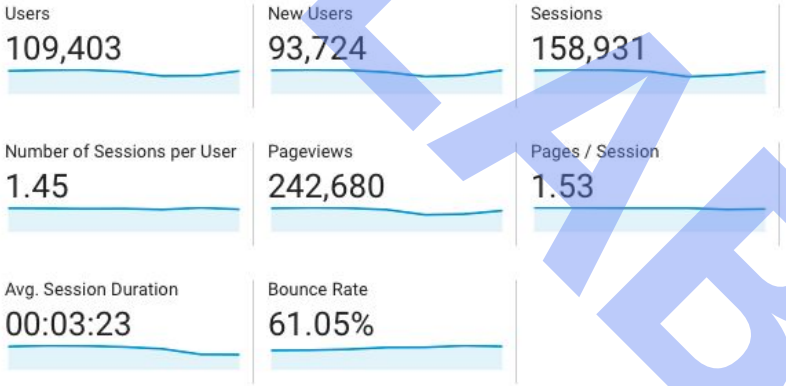
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Processing Steps - Step 4 : Perform Data Analysis - Descriptive

- Looks at what has happened in the past

Google Analytics





https://careerfoundry.com/en/blog/data-analytics/different-types-of-data-analysis/

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Processing Steps - Step 4 : Perform Data Analysis - Descriptive

- Two main techniques used in descriptive analytics:
  - **Data aggregation**: the average customer age, the average number of purchases, …
  - **Data mining**: explores the data in order to uncover any patterns or trends.
- The outcome of descriptive analysis is a **visual representation of the data**—as a bar graph, for example, or a pie chart
- Descriptive analytics condenses large volumes of data into a clear, simple overview of what has happened.

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Data Analysis

Processing Steps - Step 4 : Perform Data Analysis - Descriptive

- Statistical terms
  - Ratio
  - Proportion
  - Percentage
  - Rate
  - Mean
  - Median
  - Trend



January:	30
February:	45
March:	38
April:	41
May:	37
June:	40

30+45+38+41+37+40 = 231 clients
231 clients ÷ 6 months = 38.5

Mean = 38.5 clients/month

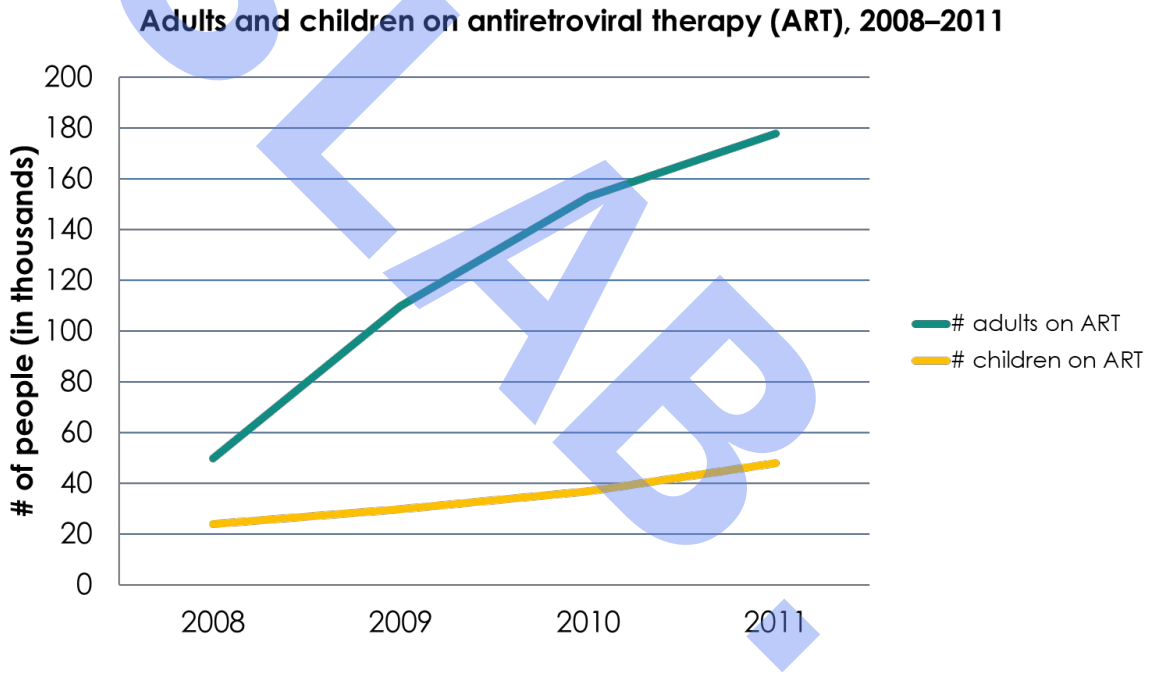
Client 1 – 2
Client 2 – 134
Client 3 – 67
Client 4 – 10
Client 5 – 221
Median of clients 1–5 = **67**
Median of clients 1–4 = **100.5**
(67+134=201/2 = 100.5)

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Processing Steps - Step 4 : Perform Data Analysis - Descriptive

Trend

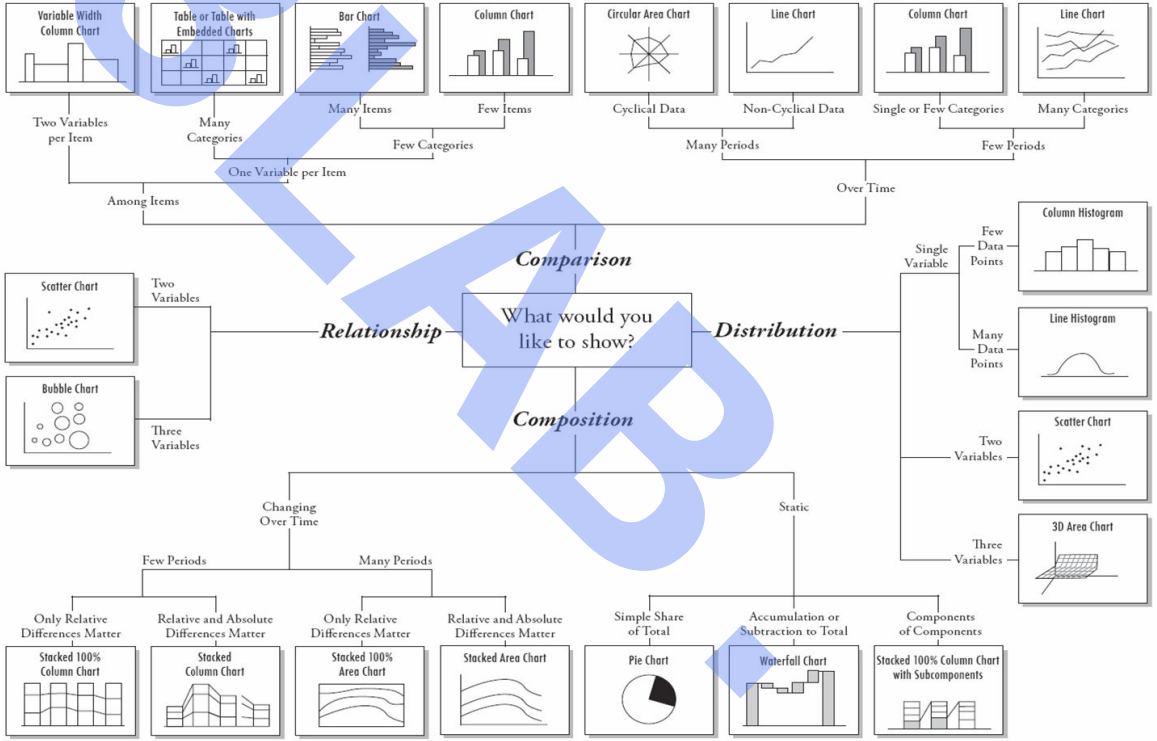


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Processing Steps - Step 4 : Perform Data Analysis - Descriptive

Types of Charts



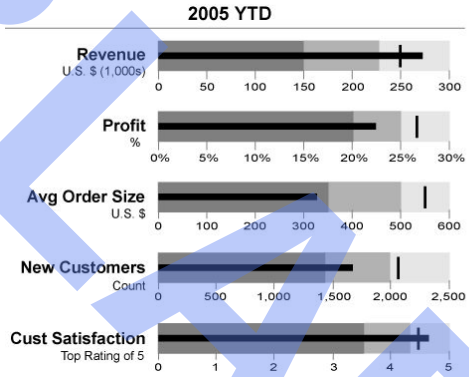
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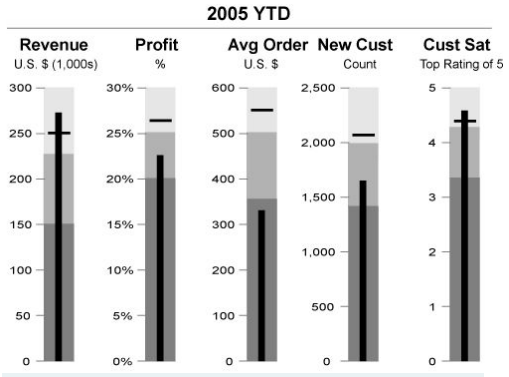
Data Analysis

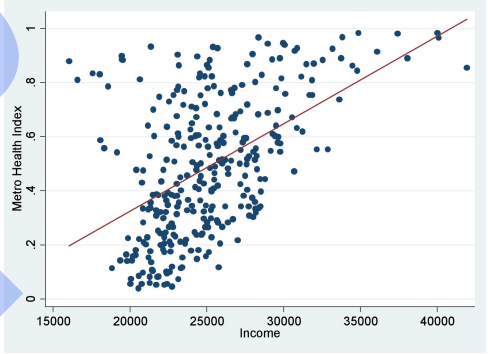
Processing Steps - Step 4 : Perform Data Analysis - Descriptive

How to choose the chart

- Want to compare values?
  - Column/bar
  - Circular area
  - Line
  - Scatter plot
  - Bullet





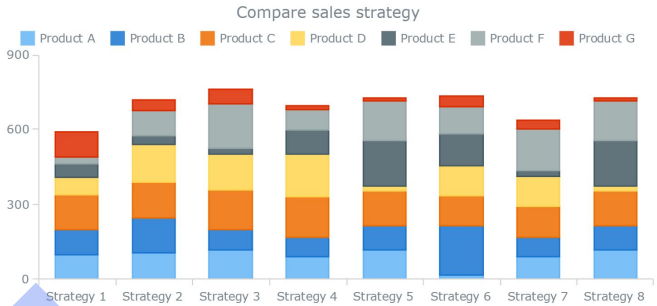


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Processing Steps - Step 4 : Perform Data Analysis - Descriptive

- Want to show the composition of something?
  - Pie
  - Stacked bar
  - Stacked column
  - Area
- Want to understand the distribution of your data?
  - Scatter plot
  - Line
  - Column
  - Bar

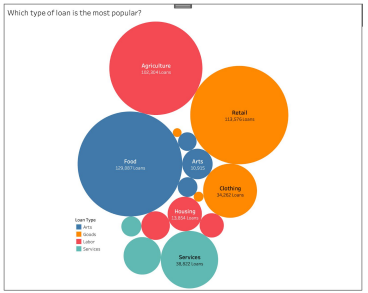


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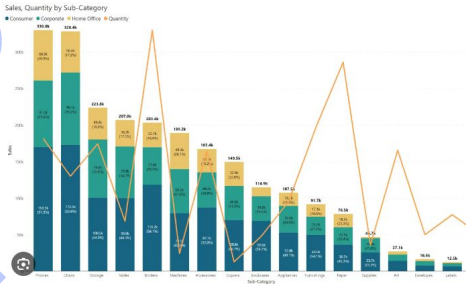
Data Analysis

Processing Steps - Step 4 : Perform Data Analysis - Descriptive

- Interested in analyzing trends in your data set?
  - Line
  - Dual-axis line
  - Column



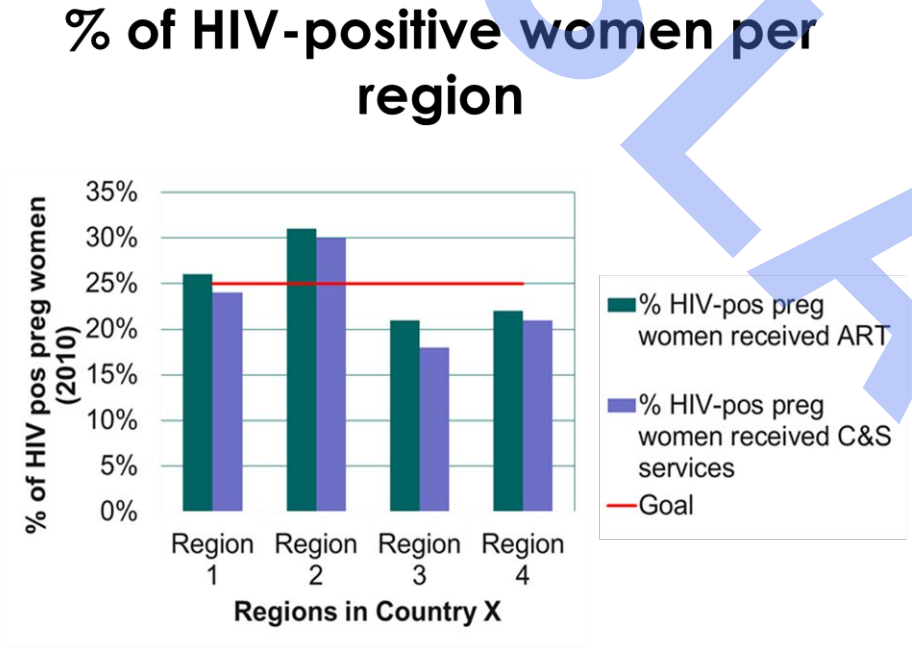
- Want to better understand the relationships among value sets?
  - Scatter plot
  - Bubble
  - Line

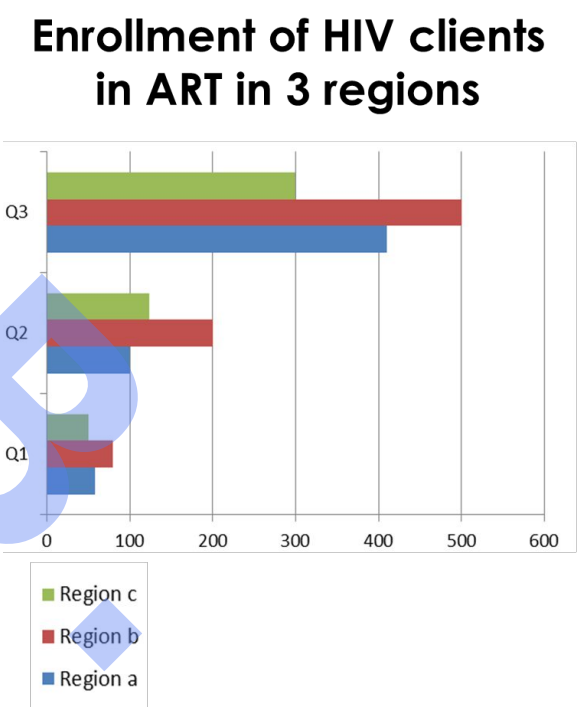


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Processing Steps - Step 4 : Perform Data Analysis - Descriptive



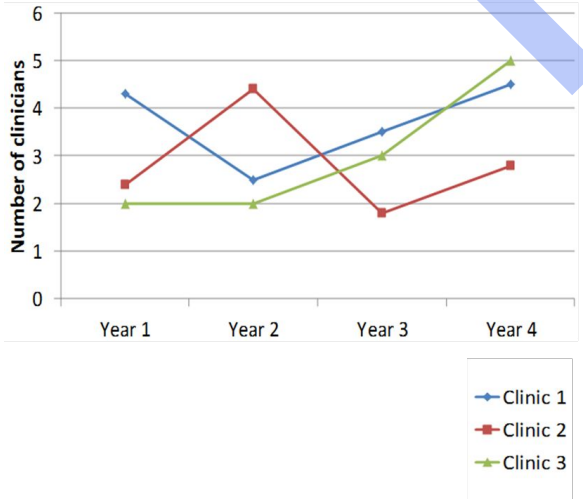


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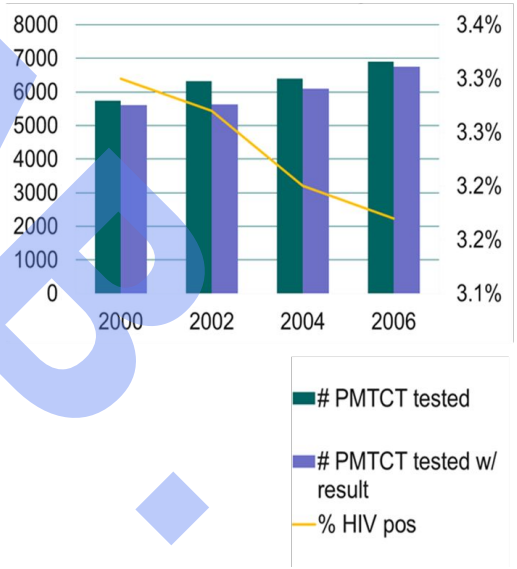
Data Analysis

Processing Steps - Step 4 : Perform Data Analysis - Descriptive

Number of clinicians working in each clinic in Years 1–4



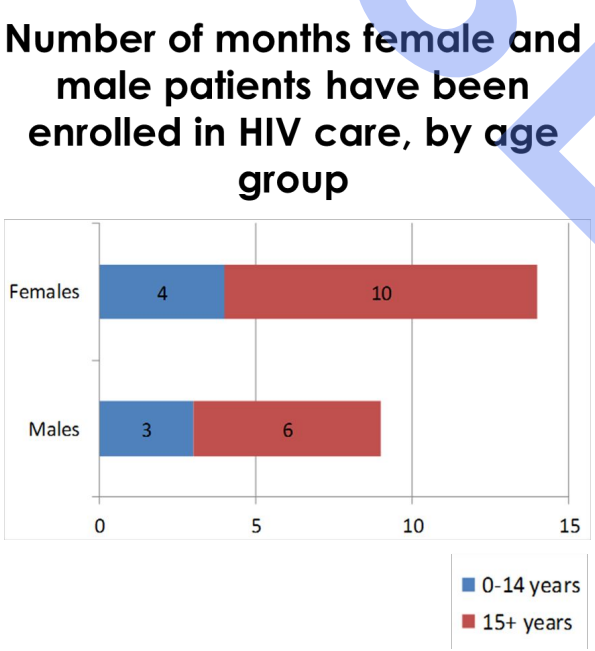
Danu Clinic
PMTCT clients tested, # received results and % with result as HIV positive, 2000-2006

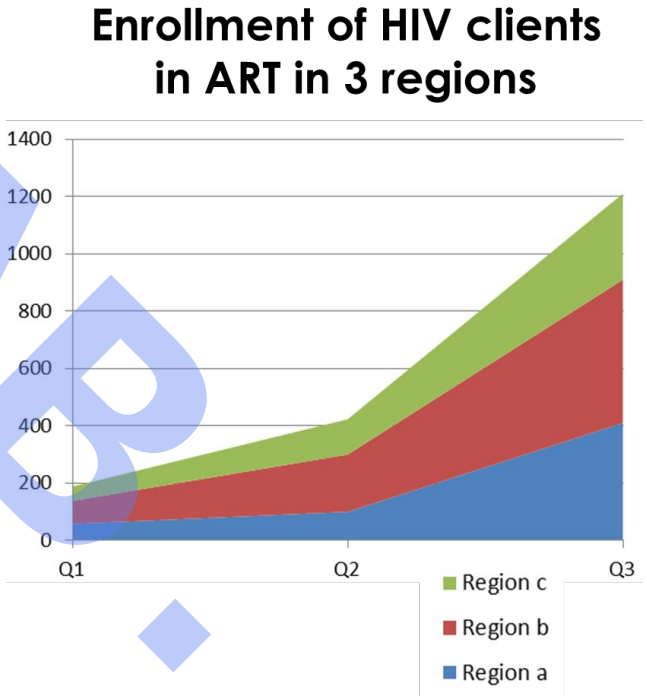


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Processing Steps - Step 4 : Perform Data Analysis - Descriptive

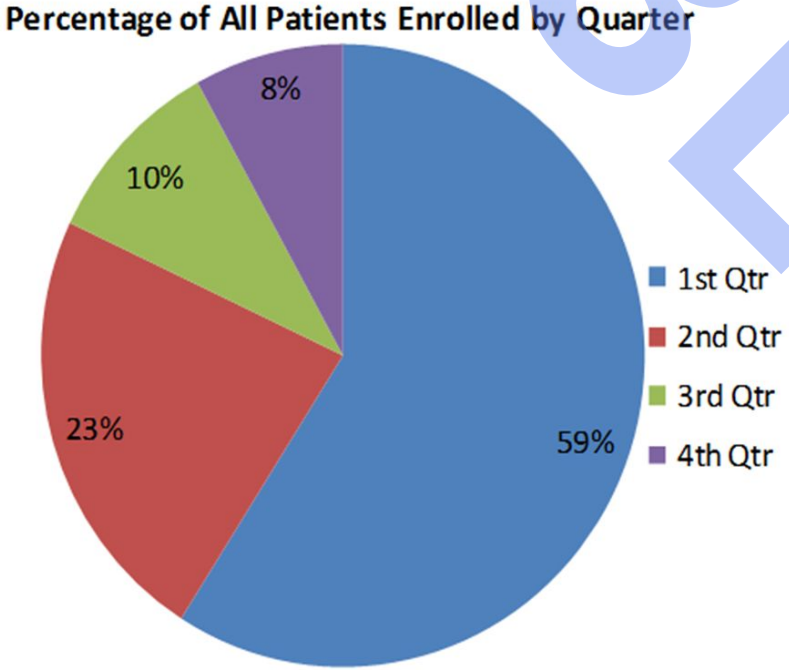


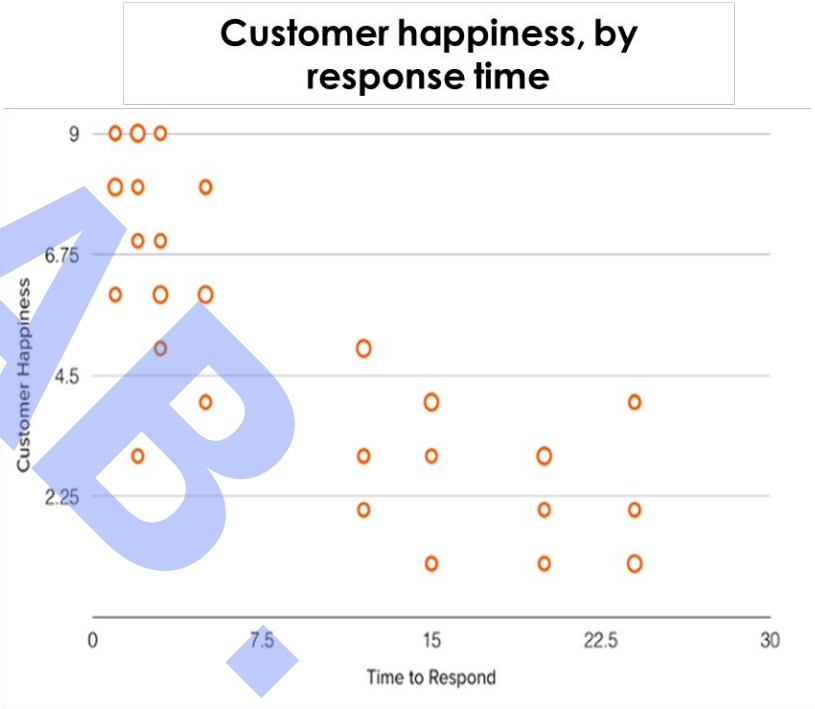


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Data Analysis

Processing Steps - Step 4 : Perform Data Analysis - Descriptive





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Data Analysis

Processing Steps - Step 4 : Perform Data Analysis - Diagnostic

- Seeks to delve deeper in order to understand why something happened
- Identify and respond to **anomalies within your data**
- Usage techniques: probability theory, regression analysis, filtering, and time-series analysis.

Why 20% drop in sales for the month of march?

website visitors, "**add to cart**" actions are increase but a few check out

-> address form problem, not work well on mobile

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Data Analysis

Processing Steps - Step 4 : Perform Data Analysis - Predictive

- Seeks to predict what is likely to happen in the **future**
- Also used for **Classification**
- It can **never** be **one hundred percent accurate**, it does eliminate much of the **guesswork**.
- **Machine learning** is a important branch of predictive analytics, designed to recognize patterns in the data and automatically evolve in order to make accurate predictions.

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Data Analysis

Processing Steps - Step 4 : Perform Data Analysis - Prescriptive

- Looks at **what has happened**, **why it happened**, and **what might happen** in order to determine what should be done next.
- The most complex type of analysis, involving algorithms, machine learning, statistical methods, and computational modeling procedures. Essentially, a prescriptive model considers all the possible decision patterns or pathways a company might take, and their likely outcomes.

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Data Analysis

Processing Steps - Step 4 : Perform Data Analysis - Prescriptive

- **Google Maps** will consider all the possible modes of transport (e.g. bus, walking, or driving), the current traffic conditions and possible roadworks in order to calculate the best route.

  "Take Route A instead of Route B to save 12 minutes."

  → This is a *prescriptive recommendation* based on traffic forecasts.

- **IBM Watson Studio / Decision Optimization** Prescriptive analytics in business & logistics, Combines machine learning (predictive) with IBM CPLEX optimization engine. Used to recommend optimal resource allocation, pricing strategies, or supply chain routes.

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Data Analysis

Processing Steps - Step 5 : Report your Findings

- You have your insights -> share these insights with the wider world (or at least with your organization's stakeholders!)
- 100% clear and unambiguous
- Use reports, dashboards, and interactive visualizations
- How to do?
  - Python Libraries: *Plotly*, *Seaborn*, and *Matplotlib* -> research, paper and reports
  - Tools: *Tableau*, *Google Studio*, *Power Bi*, *Streamlit*.
  - Javascript Library (for Developing our owned Dashboard): *Chartjs*, *Apexcharts*, *Tradingview*

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Data Analysis

Processing Steps - “Step 6” : Embrace your failures

- Data analysis is more of an iterative process than a one-way street.
- Data analytics is inherently messy, and the process you follow will be different for every project.
- Stay open-minded, and be creative.

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Time Series Analysis

Time Series (TS)

- A Time Series is a sequence of measures of a given phenomenon taken at **regular time intervals** such as hourly, daily, weekly, monthly, quarterly, annually, or every so many years.

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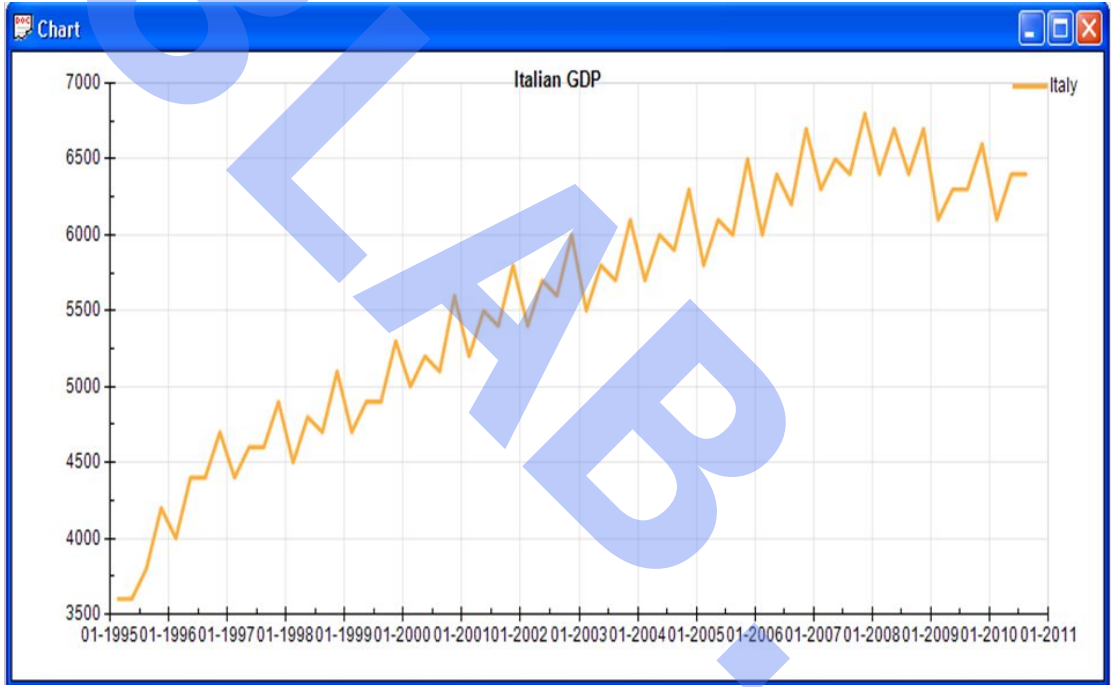
Time Series Analysis

Time Series (TS)

Italian GDP – Quarterly data

Grid

	Italy
I-1995	3600
II-1995	3600
III-1995	3800
IV-19...	4200
I-1996	4000
II-1996	4400
III-1996	4400
IV-19...	4700
I-1997	4400
II-1997	4600
III-1997	4600
IV-19...	4900
I-1998	4500
II-1998	4800
III-1998	4700
IV-19...	5100
I-1999	4700
II-1999	4900



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Time Series Analysis

Time Series (TS)

2008				2009			
Q1	Q2	M8	Q4	Q1	Q2	Q3	Q4
100	200	30	250	90	120	100	190

2008				2009			
Q1	Q2		Q4	Q1	Q2	Q3	Q4
100	200		250	90	120	100	190

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Time Series Analysis

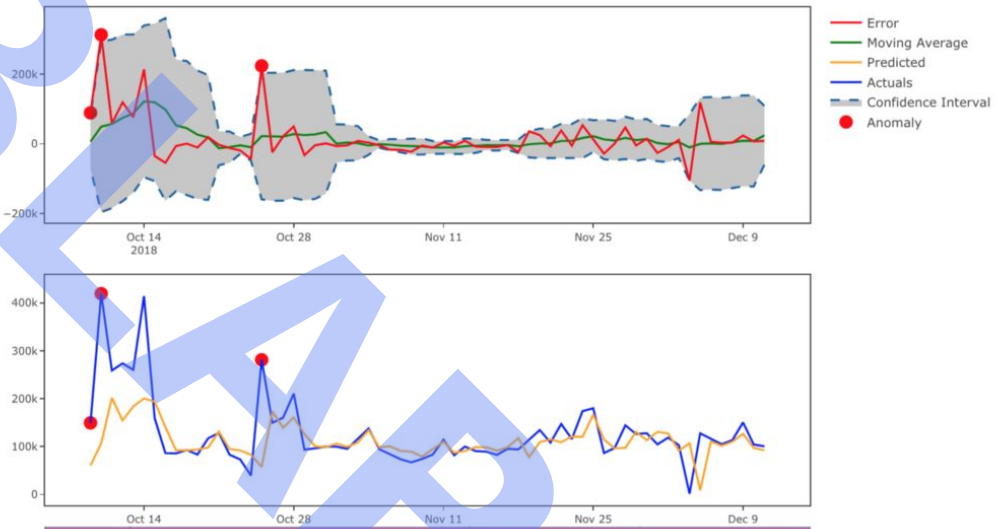
Time Series Analysis (TSA)

- **Describe**: Describe the important features of the time series data. The first step is to **plot the data** to look for the possible presence of trends, seasonal variations, outliers and turning points.
- **Model**: Investigate and find out the generating process of the time series.
- **Predict**: Forecast future values of an observed time series. Applications are in predicting stock prices or product sales.

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Time Series Analysis

Time Series Analysis (TSA)



Date	Actual Values	Predicted	% Difference	Severity (0-3)
2018-11-22	147069	108739.168	26.062%	0
2018-11-21	108126	114978.294	-6.337%	0
2018-11-20	134514	109228.065	18.798%	0
2018-11-19	113630	77791.011	31.54%	2
2018-11-18	94102	117629.725	-25.002%	1
2018-11-17	95410	98640.56	-3.386%	0

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Time Series Analysis

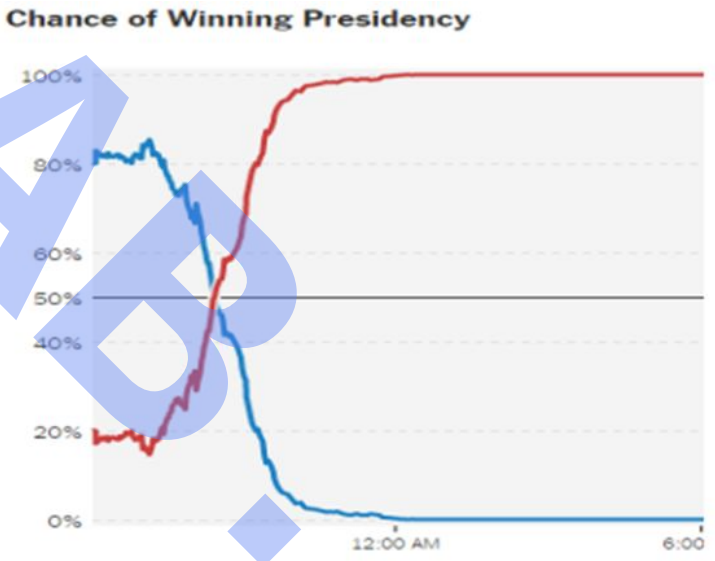
Time Series Analysis (TSA)

- Forecasting is the process of making predictions of the future based on past and present data.

College Station, TX 10 Day Weather

7:51 am CDT Print

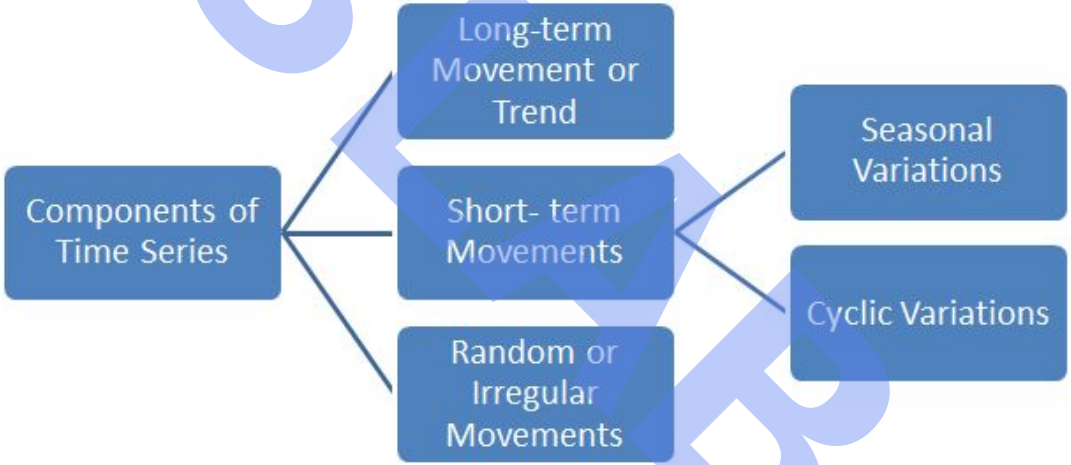
DAY	DESCRIPTION	HIGH / LOW	PRECIP	WIND	HUMIDITY
TODAY APR 13	Thunderstorms/Wind	80/57°	80%	S 20 mph	82%
SAT APR 14	Partly Cloudy	66/41°	0%	NNW 18 mph	41%
SUN APR 15	Sunny	68/44°	0%	NNW 17 mph	39%
MON APR 16	Sunny	76/52°	0%	SSE 8 mph	44%
TUE APR 17	Sunny	81/62°	10%	S 18 mph	57%
WED APR 18	Mostly Cloudy	81/59°	10%	SW 9 mph	69%
THU APR 19	AM Clouds/PM Sun	80/60°	0%	E 10 mph	55%
FRI APR 20	Cloudy	78/64°	20%	SE 18 mph	65%
SAT APR 21	Scattered Thunderstorms	77/55°	50%	SSW 13 mph	75%
SUN	Sunny	77/56°	10%	N 10 mph	51%



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Time Series Analysis

Key concepts - Components



$$Y_t = T_t + S_t + C_t + I_t$$

Time Series Analysis

Key concepts - Components

	Trend	Seasonality	Cyclical	Irregularity
Time	Fixed Time Interval	Fixed Time Interval	Not Fixed Time Interval	Not Fixed Time Interval
Duration	Long and Short Term	Short Term	Long and Short Term	Regular/Irregular
Visualization				
Nature - I	Gradual	Swings between Up or Down	Repeating Up and Down	Errored or High Fluctuation
Nature – II	Upward/Down Trend	Pattern repeatable	No fixed period	Short and Not repeatable
Prediction Capability	Predictable	Predictable	Challenging	Challenging
Market Model				Highly random/Unforeseen Events – along with white noise.

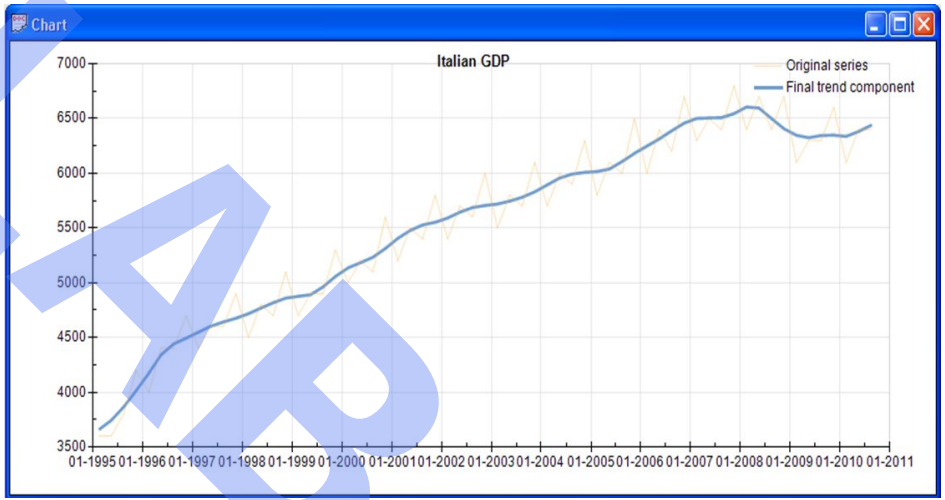
Designed by Author (Shanthababu)

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Time Series Analysis

Key concepts - Components - Trend

- A Trend is exists when there is a long-term increase or decrease in the data. It doesn't have to be linear. Sometimes we will refer to a trend as "changing direction" when it goes from an increasing trend to a decreasing trend.

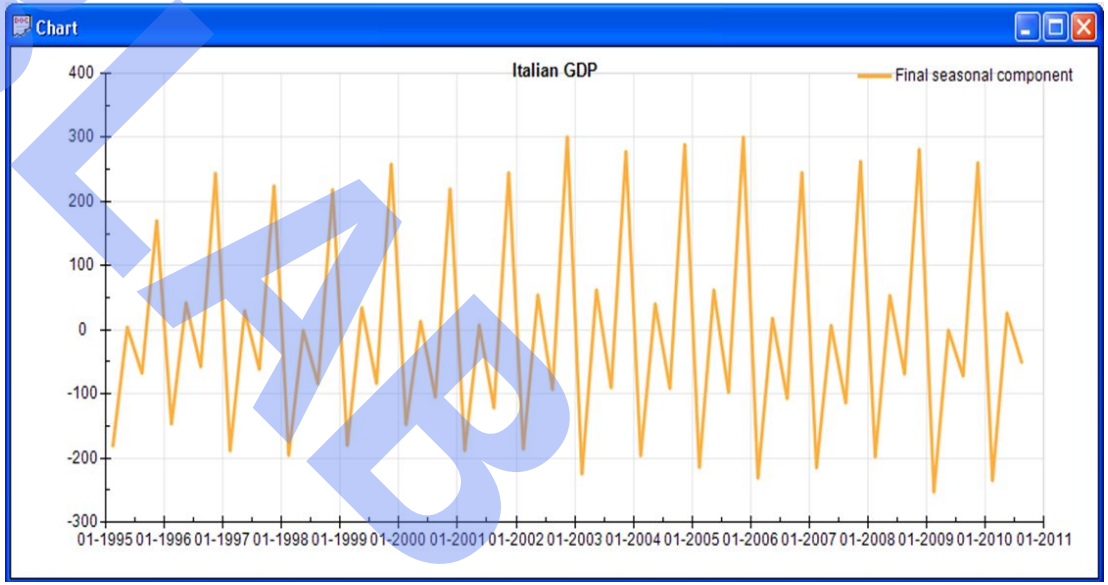


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Time Series Analysis

Key concepts - Components - Seasonality

- A seasonal pattern exists when a series is influenced by seasonal factors (quarterly, monthly, half-yearly). Seasonality is always of a fixed and known period.



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Time Series Analysis

Key concepts - Components - Cyclic Variations

- A cyclic pattern exists when data exhibits rises and falls that are not of fixed period. The duration of these cycles is more than a year. For example, stock prices cycle between periods of high and low values but there's no set amount of time between those fluctuations.

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Time Series Analysis

Key concepts - Components - Irregular or Random Walk

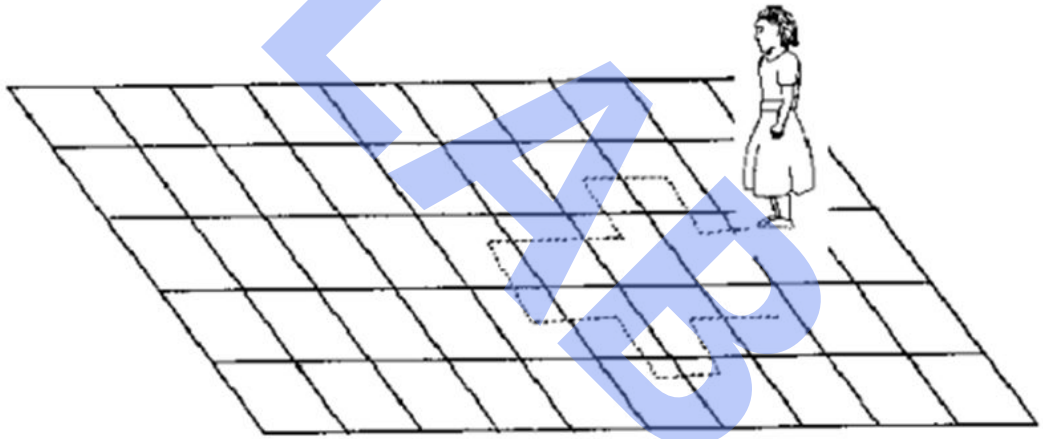
- The variation of observations in a time series which is unusual or unexpected. It's also termed as a Random Variation and is usually unpredictable. Floods, fires, revolutions, epidemics, and strikes are some examples

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Time Series Analysis

Key concepts - Components - Irregular or Random Walk

$$X(t) = X(t-1) + Er(t)$$

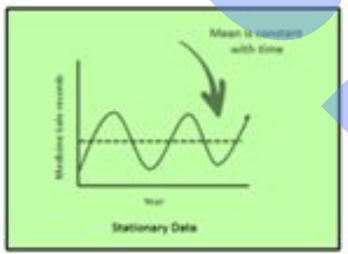
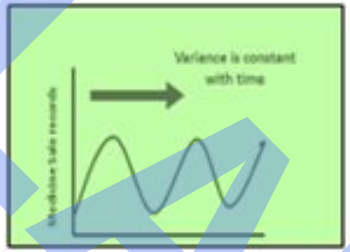
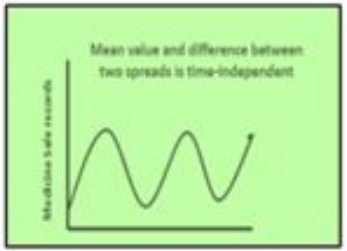
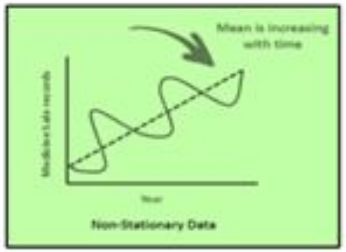
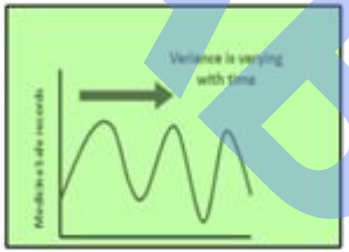
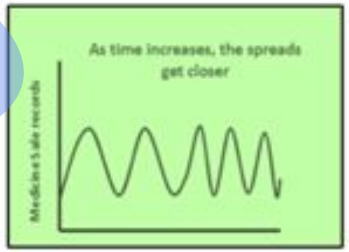


(Source: http://scifun.chem.wisc.edu/WOP/RandomWalk.html)

S3LAB

Time Series Analysis

Key concepts - Data type

	MEAN	Variance	Covariance
Stationary			
Non-Stationary			

Designed by Author (Shanthababu)

SSLAB

Time Series Analysis

Key concepts - Data type

- Given a series of data points, if the **mean** and **variance** of all the data points remain constant with time, then we call it a **stationary series**. If these **vary with time**, we call it a **non-stationary series**.
- Most prices (such as stock prices or price of Bitcoins) are not stationary. They are either drifting upward or downward. **Non-stationary data are unpredictable and cannot be modeled or forecasted**. The results obtained by using non-stationary time series may be spurious in that they may indicate a relationship between two variables where one doesn't exist. In order to receive consistent, reliable results, non-stationary data needs to be transformed into stationary data

SSTAB

Time Series Analysis

Key concepts - Data type - How to make it Stationary

- **Differencing**: In order to make a series stationary, we take a difference between the data points. Suppose the original time series is $X_1, X_2, X_3, \ldots X_n$. Series with difference of degree 1 becomes $X_2 - X_1, X_3 - X_2, X_4 - X_3, \ldots, X_n - X_{n-1}$. If this transformation is done only once to a series, we say that the data has been first differenced. This process essentially eliminates the trend if the series is growing at a fairly constant rate. If it's growing at an increasing/decreasing rate, we can apply the same procedure and difference the data again. The data would then be second differenced.
- **Transformation**: If the series can't be made stationary, we can try transforming the variables. **Log transform** is probably the most commonly used transformation for a diverging time series. However, it's normally suggested to use transformation only when differencing is not working.

SCALED

Stock Data Analysis

Understand about the Business

- Read book and tutorial about the Stock:

  -https://www.vietcap.com.vn/kien-thuc/kien-thuc-co-ban-ve-chung-khoan-nha-dau-tu-nhat-dinh-phai-biet

  -https://www.ssi.com.vn/khach-hang-ca-nhan/trung-tam-kien-thuc

  -https://www.vietcap.com.vn/kien-thuc/top-5-cuon-sach-dau-tu-chung-khoan-co-ban-cho-nguoi-moi-bat-dau

S3T AB.

Stock Data Analysis

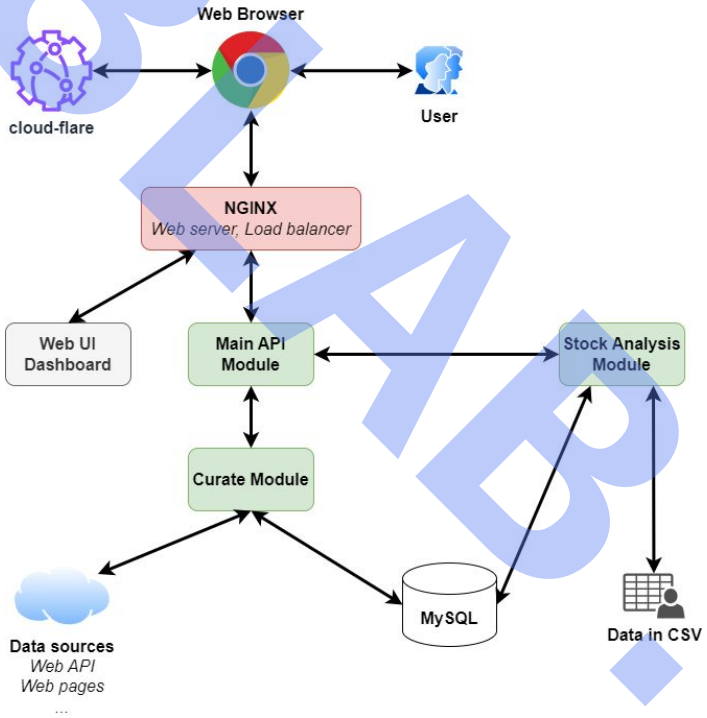
Define Your Objective - Overview

- Visualize the Data, making different views about Stock Data
- Predict future stock prices using machine learning models
- Building an Automated Trading System using Deep Reinforcement Learning
  - Make autonomous BUY/SELL/HOLD decisions based on predictions
  - Maximize portfolio returns by learning from market behavior
- Adapt trading strategy to changing market conditions

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Stock Data Analysis

Define Your Objective - System Architecture



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Stock Data Analysis

Define Your Objective - Dashboard



SSLAB

Stock Data Analysis

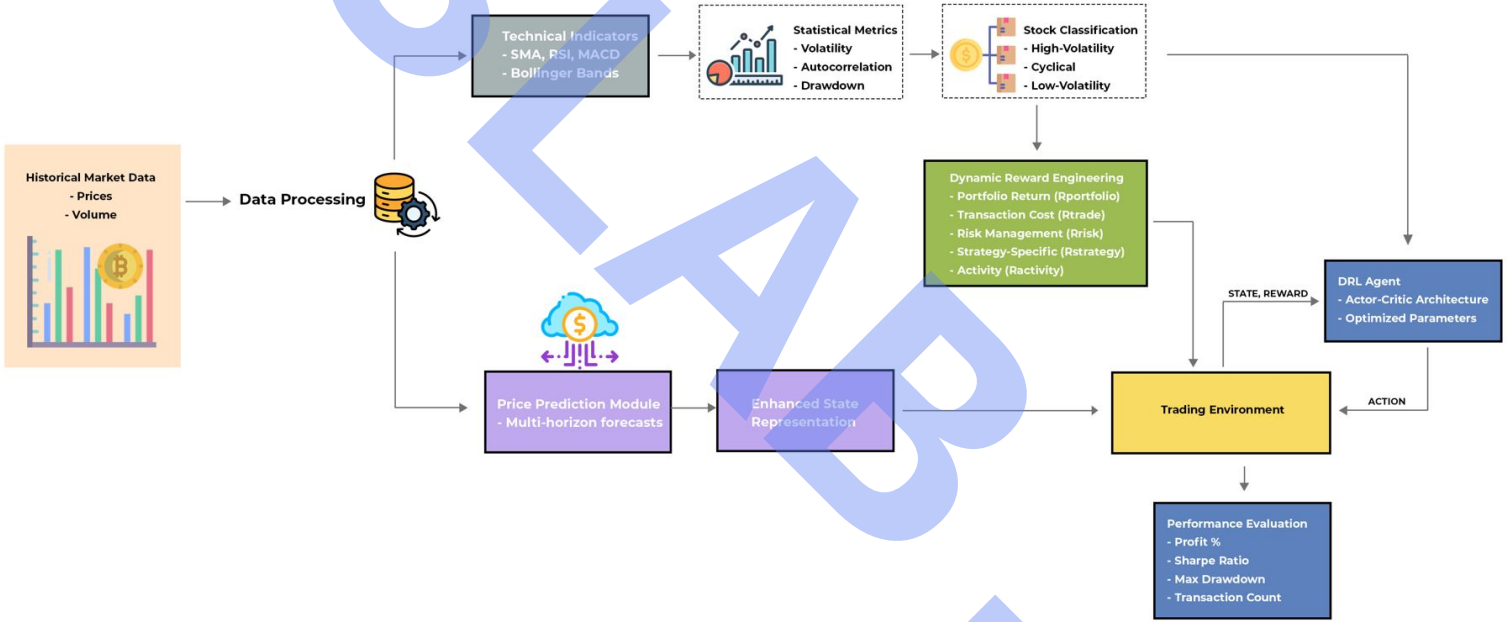
Define Your Objective - Dashboard



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Stock Data Analysis

General Processing Flow



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Stock Data Analysis

Collect the Data

- Using Web API
- Using Web scraping

We got the market, industry, stock prices information by scraping the page:

https://dstock.vndirect.com.vn

https://cafef.vn/

Or Call the Web API

https://api.vietstock.vn

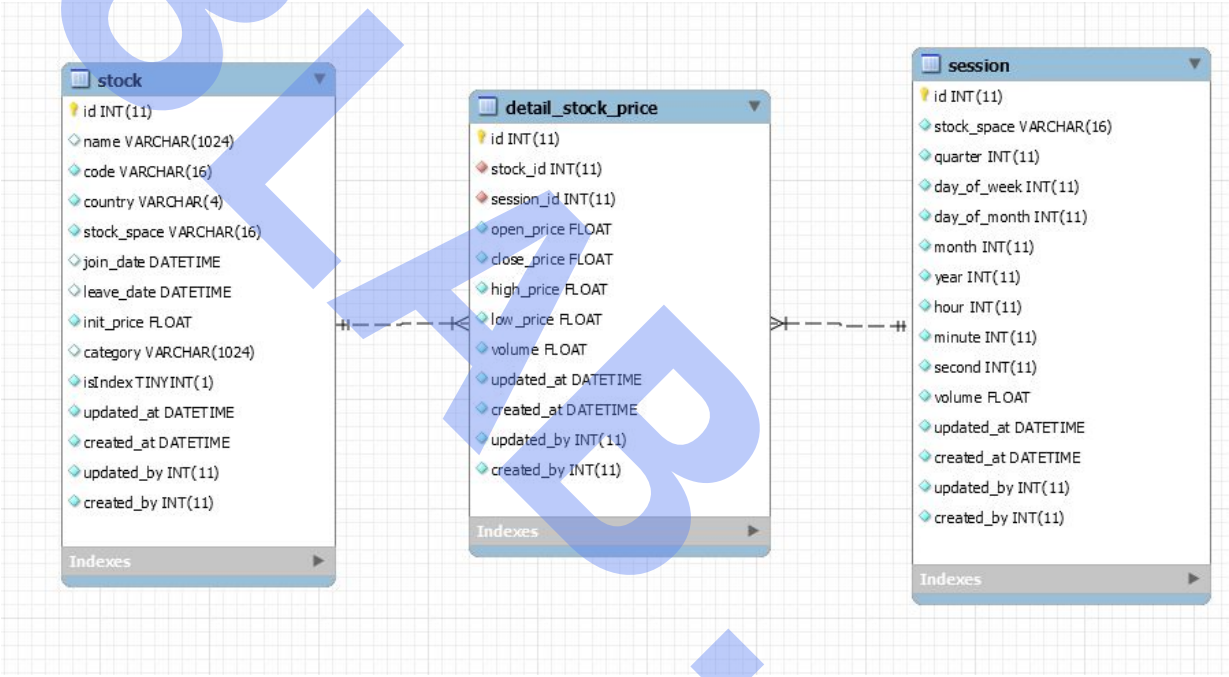
Using: Postman, Python for test and collect Data

S3L.AB.

Stock Data Analysis

Clean the Data

Make Data
In Structure



S3LAB.

Stock Data Analysis

Clean the Data

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: | Fetch rows:

id	stock_id	session_id	open_price	close_price	high_price	low_price	volume	updated_at	created_at	updated_by	created_by
1	1	1	1280.26	1266.91	1283.06	1263.87	1139090000	2024-05-23 03:17:49	2024-05-23 03:17:49	1	1
2	1	2	1277.85	1277.14	1281.29	126	72160000	2024-05-23 03:17:49	2024-05-23 03:17:49	1	1
3	1	3	1281.5	1277.58	1285.19	1277.36	1161560000	2024-05-23 03:17:49	2024-05-23 03:17:49	1	1
4	1	4	1270.34	1273.11	1274.27	1266.32	937043000	2024-05-23 03:17:49	2024-05-23 03:17:49	1	1
5	1	5	1264.51	1268.78	1271.66	1261.19	885056000	2024-05-23 03:17:49	2024-05-23 03:17:49	1	1
6	1	6	1248.61	1254.39	1257.92	1248.02	880174000	2024-05-23 03:17:49	2024-05-23 03:17:49	1	1
7	1	7	1246.23	1243.28	1252.52	1240.88	640494000	2024-05-23 03:17:49	2024-05-23 03:17:49	1	1
8	1	8	1248.19	1240.18	1249.2	1233.56	741579000	2024-05-23 03:17:49	2024-05-23 03:17:49	1	1
9	1	9	1251.78	1244.7	1252.5	1237.46	669111000	2024-05-23 03:17:49	2024-05-23 03:17:49	1	1
10	1	10	1252.81	1248.64	1256.8	1243.29	771553000	2024-05-23 03:17:49	2024-05-23 03:17:49	1	1
11	1	11	1243.61	1250.46	1251.62	1234.76	981803000	2024-05-23 03:17:49	2024-05-23 03:17:49	1	1
12	1	12	1245.5	1248.63	1250.63	1240.09	707897000	2024-05-23 03:17:49	2024-05-23 03:17:49	1	1
13	1	13	1226.89	1241.58	1241.58	1224.22	864634000	2024-05-23 03:17:49	2024-05-23 03:17:49	1	1
14	1	14	1222.54	1221.03	1225.75	1218.33	643429000	2024-05-23 03:17:49	2024-05-23 03:17:49	1	1
15	1	15	1212.13	1216.36	1216.36	1206.28	570364000	2024-05-23 03:17:49	2024-05-23 03:17:49	1	1

SESI LAB

Stock Data Analysis

Clean the Data

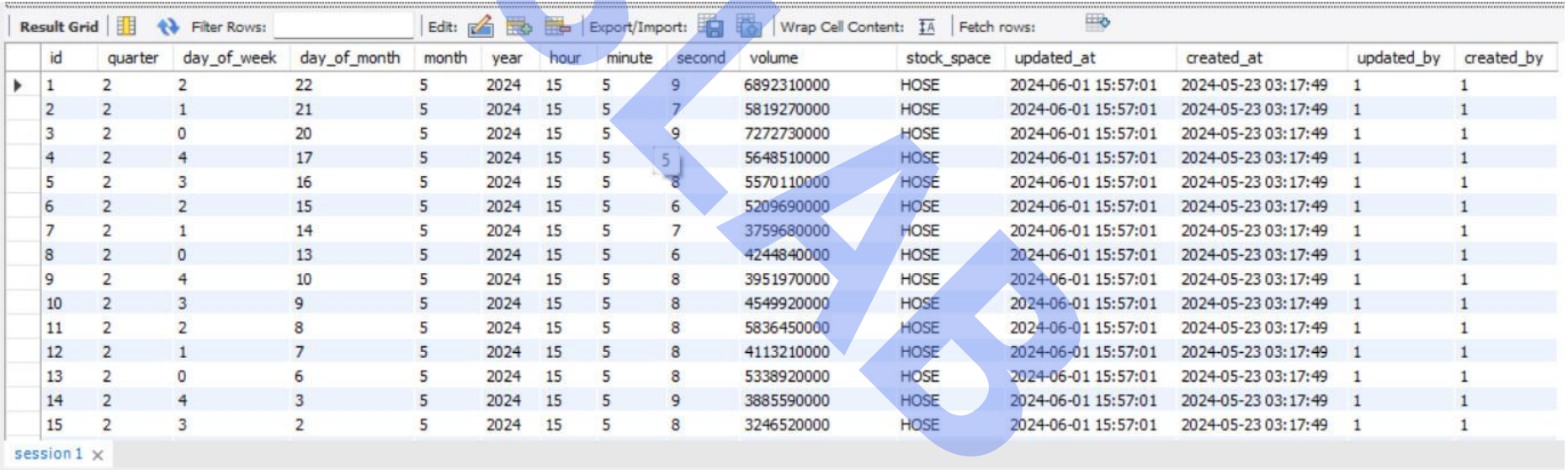
id	name	code	country	stock_space	join_date	leave_date	init_price	updated_at	created_at	updated_by	created_by	category	isIndex
182	Công ty Cổ p...	TQN	VI	UPCOM	2017-02-08 00:00:00	NULL	12894.7	2024-05-23 03:11:51	2024-05-23 03:11:51	1	1	Tài nguyên	0
183	Công ty Cổ p...	TNB	VI	UPCOM	2010-04-16 00:00:00	NULL	7064.15	2024-05-23 03:11:51	2024-05-23 03:11:51	1	1	Tài nguyên	0
184	Công ty Cổ p...	KSH	VI	UPCOM	2019-09-19 00:00:00	NULL	16546.1	2024-05-23 03:11:51	2024-05-23 03:11:51	1	1	Tài nguyên	0
185	Công ty Cổ p...	CKD	VI	UPCOM	2015-05-21 00:00:00	NULL	4937.96	2024-05-23 03:11:51	2024-05-23 03:11:51	1	1	Tài nguyên	0
186	Công ty Cổ p...	KSB	VI	HOSE	2010-01-20 00:00:00	NULL	2723.49	2024-05-23 03:11:51	2024-05-23 03:11:51	1	1	Tài nguyên	0
187	Công ty Cổ p...	KCB	VI	UPCOM	2015-12-01 00:00:00	NULL	9400	2024-05-23 03:11:51	2024-05-23 03:11:51	1	1	Tài nguyên	0
188	Công ty cổ p...	CKA	VI	UPCOM	2018-10-17 00:00:00	NULL	13693.9	2024-05-23 03:11:51	2024-05-23 03:11:51	1	1	Tài nguyên	0
189	Công ty cổ p...	PLP	VI	HOSE	2017-08-21 00:00:00	NULL	7164.96	2024-05-23 03:11:51	2024-05-23 03:11:51	1	1	Tài nguyên	0
190	Công ty Cổ p...	THT	VI	HNX	2008-10-24 00:00:00	NULL	2130.98	2024-05-23 03:11:51	2024-05-23 03:11:51	1	1	Tài nguyên	0
191	Công ty cổ p...	VDB	VI	UPCOM	2018-11-22 00:00:00	NULL	900.001	2024-05-23 03:11:51	2024-05-23 03:11:51	1	1	Tài nguyên	0
192	Công ty Cổ p...	HPG	VI	HOSE	2007-11-15 00:00:00	NULL	2728.96	2024-05-23 03:11:51	2024-05-23 03:11:51	1	1	Tài nguyên	0
193	Công ty Cổ p...	TEG	VI	HOSE	2018-03-09 00:00:00	NULL	6868.35	2024-05-23 03:11:51	2024-05-23 03:11:51	1	1	Tài nguyên	0
194	CÔNG TY CỔ ...	KLM	VI	UPCOM	2019-12-30 00:00:00	NULL	10400	2024-05-23 03:11:51	2024-05-23 03:11:51	1	1	Tài nguyên	0
195	Công ty Cổ p...	MDC	VI	HNX	2009-07-23 00:00:00	NULL	3193.15	2024-05-23 03:11:51	2024-05-23 03:11:51	1	1	Tài nguyên	0

Result Grid
Form Editor
Field Types
Query Stats

S3IT B.

Stock Data Analysis

Clean the Data



Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: | Fetch rows:

id	quarter	day_of_week	day_of_month	month	year	hour	minute	second	volume	stock_space	updated_at	created_at	updated_by	created_by
1	2	2	22	5	2024	15	5	9	6892310000	HOSE	2024-06-01 15:57:01	2024-05-23 03:17:49	1	1
2	2	1	21	5	2024	15	5	7	5819270000	HOSE	2024-06-01 15:57:01	2024-05-23 03:17:49	1	1
3	2	0	20	5	2024	15	5	9	7272730000	HOSE	2024-06-01 15:57:01	2024-05-23 03:17:49	1	1
4	2	4	17	5	2024	15	5	5	5648510000	HOSE	2024-06-01 15:57:01	2024-05-23 03:17:49	1	1
5	2	3	16	5	2024	15	5	8	5570110000	HOSE	2024-06-01 15:57:01	2024-05-23 03:17:49	1	1
6	2	2	15	5	2024	15	5	6	5209690000	HOSE	2024-06-01 15:57:01	2024-05-23 03:17:49	1	1
7	2	1	14	5	2024	15	5	7	3759680000	HOSE	2024-06-01 15:57:01	2024-05-23 03:17:49	1	1
8	2	0	13	5	2024	15	5	6	4244840000	HOSE	2024-06-01 15:57:01	2024-05-23 03:17:49	1	1
9	2	4	10	5	2024	15	5	8	3951970000	HOSE	2024-06-01 15:57:01	2024-05-23 03:17:49	1	1
10	2	3	9	5	2024	15	5	8	4549920000	HOSE	2024-06-01 15:57:01	2024-05-23 03:17:49	1	1
11	2	2	8	5	2024	15	5	8	5836450000	HOSE	2024-06-01 15:57:01	2024-05-23 03:17:49	1	1
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14	2	4	3	5	2024	15	5	9	3885590000	HOSE	2024-06-01 15:57:01	2024-05-23 03:17:49	1	1
15	2	3	2	5	2024	15	5	8	3246520000	HOSE	2024-06-01 15:57:01	2024-05-23 03:17:49	1	1

session 1

S3T.AI

Stock Data Analysis

Analyze the Data - Descriptive

Using
Power BI



S3LAB

Stock Data Analysis

Analyze the Data - Descriptive



S3L-AB-1

Stock Data Analysis

Analyze the Data - Descriptive



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Stock Data Analysis

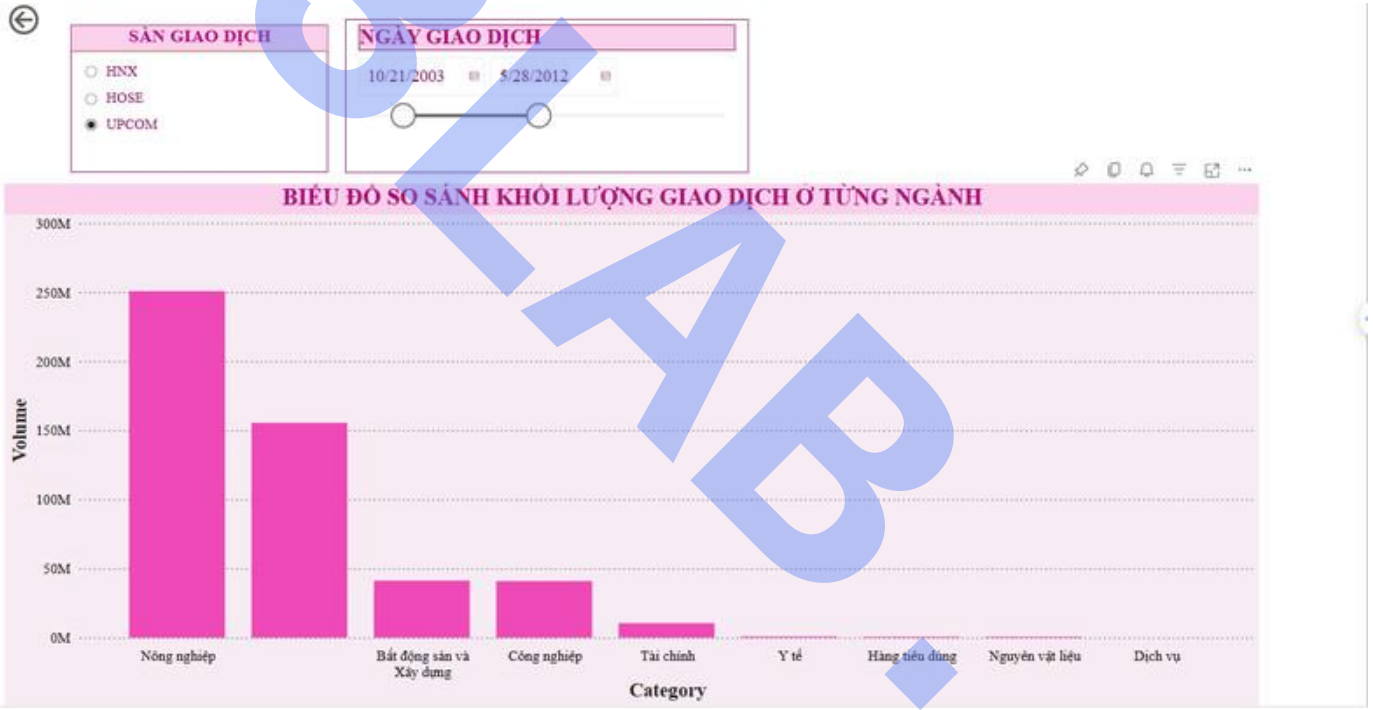
Analyze the Data - Descriptive



SSLAB

Stock Data Analysis

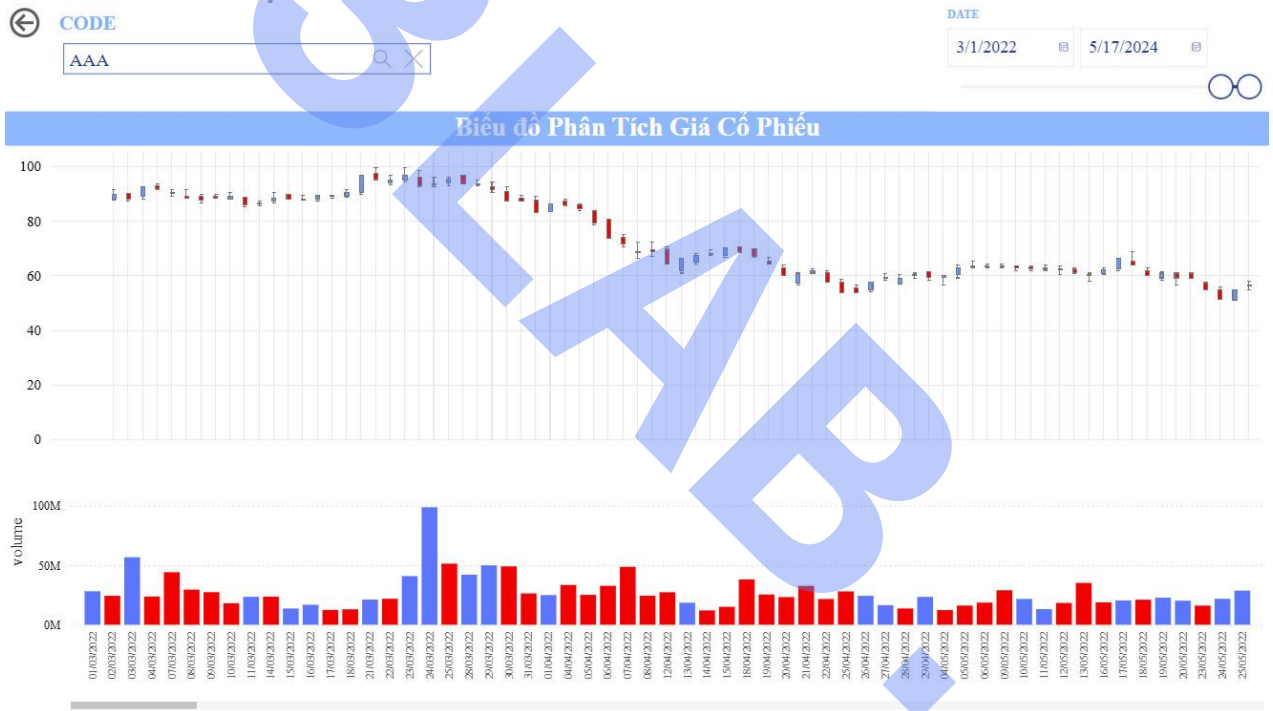
Analyze the Data - Descriptive



SSL-AB

Stock Data Analysis

Analyze the Data - Descriptive



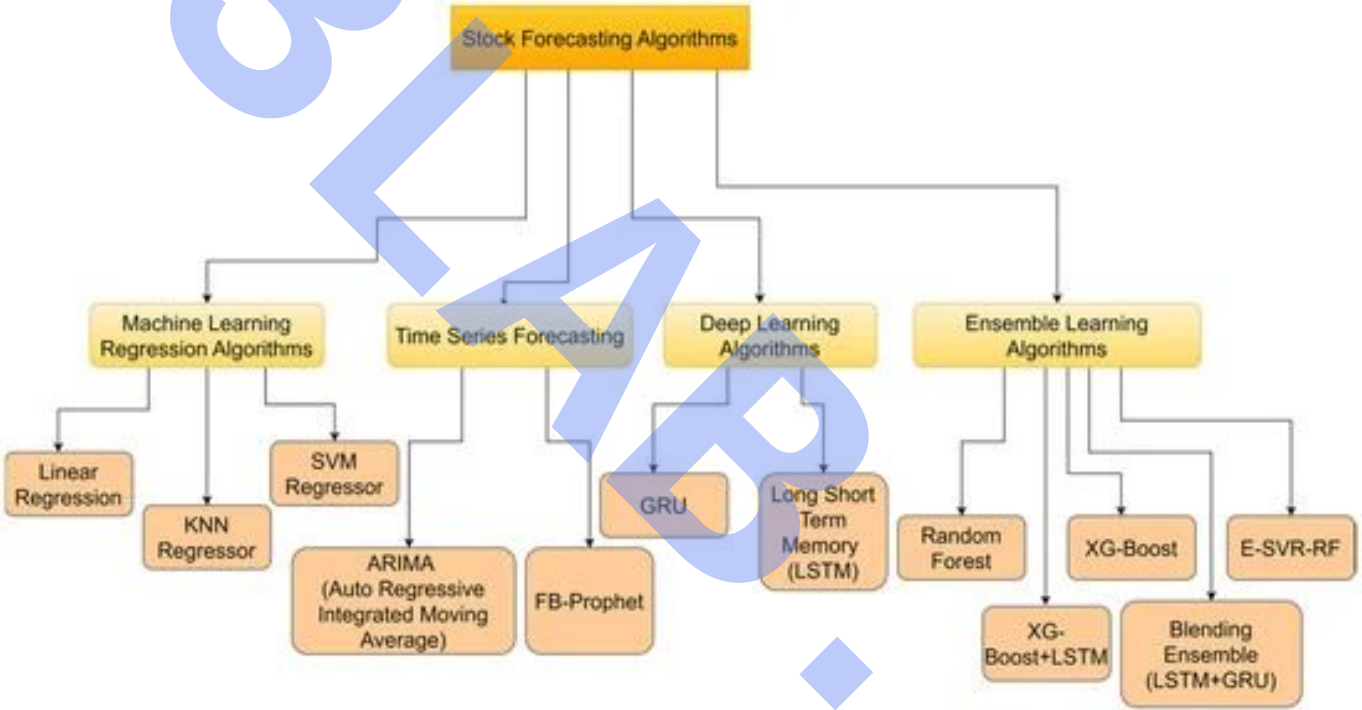
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Stock Data Analysis

Analyze the Data - Predictive

Google Scholar,
Survey Papers

Ask Goolge, Gemini, or chatGPT to look for an Insight about Stock Forecasting



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Stock Data Analysis

Analyze the Data - Predictive

- Stock Price Forecasting with Deep Learning
  - Predict future prices: 1-day, 7-day, 14-day ahead
  - Use historical patterns to understand market trends
- Why prediction is important
  - Better timing for buy/sell actions
  - Agent needs future information to make smart decisions
  - Reduces risk in uncertain markets

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Stock Data Analysis

Proposed Model Architecture (MST-Net)

Our approach:

"MST-Net: A Multi-Scale Transformer Network with Price-based Sinusoidal Encoding for Vietnam Stock Price Prediction"

- A novel hybrid deep learning model for stock price forecasting
- Captures both short-term and long-term market dynamics
- Tailored for Vietnamese stock market data (2020–2025)

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Stock Data Analysis

Proposed Model Architecture (MST-Net)

- Stock price forecasting is a core challenge in computational finance
- Traditional models (**ARIMA**, **GARCH**) assume linearity and stationarity
- Deep learning models (**LSTM**, **GRU**) capture temporal patterns but miss long-range and multi-scale dynamics
- The Latest hybrid models: News + History Stock Price -> not stable
- Vietnamese stock market: highly volatile, data-scarce, and retail-dominated

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Stock Data Analysis

Proposed Model Architecture (MST-Net)

Model Type	Examples	Strengths	Limitations
Statistical	ARIMA, GARCH	Simple, interpretable	Assume linearity, fail with non-stationary data
Machine Learning	SVM, Random Forest	Handle non-linearity	No temporal memory
Deep Learning	LSTM, GRU	Capture short-term dependencies	Struggle with long-range & multi-scale patterns
Transformer-based	Transformer, Time2Vector	Learn long dependencies	Ignore price-level cycles & local patterns

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Stock Data Analysis

Proposed Model Architecture (MST-Net)

- Existing models fail to capture multi-scale temporal patterns in stock data
- Price-level cyclic behaviors are not effectively represented
- Most studies focus on mature markets (U.S., China, etc.), not emerging ones

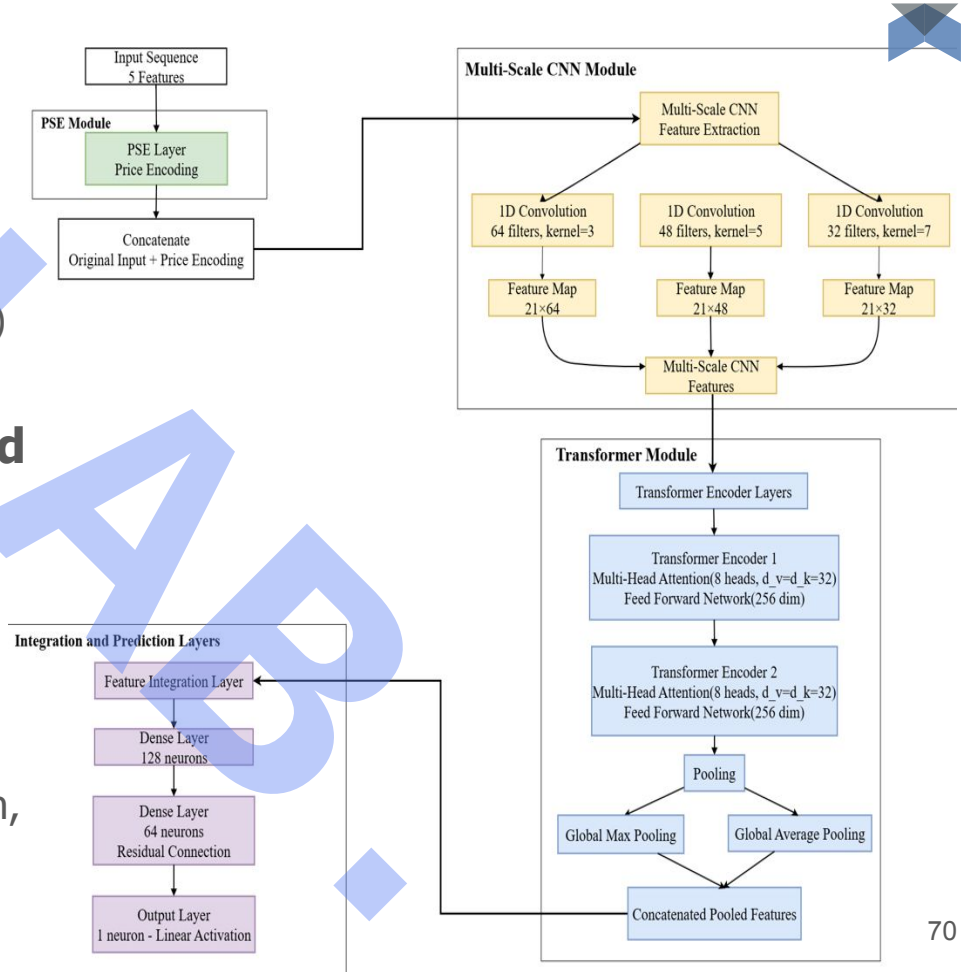
→ How can we design a model that captures both multi-scale and price-level patterns, especially for the Vietnamese market?

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Stock Data Analysis

Analyze the Data - Predictive

- Uses 21-day historical data
  (Open, High, Low, Close prices and Volume)
- Combines: **Transformer**, **Multi-Scale CNN**, and **Price-based Sinusoidal Encoding (PSE)**
- Tested on 9 Vietnamese stocks
  - **VCB**, **VNM**, **PLX**, **HPG**, **VIC**,**VHM**, **TCB**, **DPM**, **KDC**
  - From 2020 to 2025: 16055 samples (11239 for training, 2408 for validation, 2408 for test)



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Stock Data Analysis

Analyze the Data - Predictive

MST-Net outperforms all base models across 1-day, 7-day, and 14-day predictions

Tuning, Evaluate and Compare by using metrics:
RMSE, MAE, MAPE, R^2, DA

Model	Horizon	RMSE	MAE	MAPE (%)	R^2	DA (%)
Stacked LSTM	1 day	2.24	1.66	4.02	0.31	43.77
	7 days	2.93	2.21	5.93	-0.16	43.62
	14 days	3.56	2.79	6.75	-2.51	44.90
GRU	1 day	3.03	2.68	6.27	-5.91	44.44
	7 days	4.53	3.77	8.96	-9.92	42.78
	14 days	5.16	4.14	10.69	-14.69	41.75
CNN-BiLSTM-Attention	1 day	2.42	1.99	4.57	-0.19	42.86
	7 days	3.09	2.42	5.71	-0.77	40.46
	14 days	3.56	2.76	6.69	-1.09	43.06
Transformer	1 day	2.23	1.74	3.88	0.21	44.17
	7 days	3.05	2.32	5.41	-0.26	43.17
	14 days	3.50	2.65	6.61	-1.39	44.66
Transformer-CNN	1 day	2.17	1.62	3.98	0.25	42.24
	7 days	2.92	2.16	5.39	-0.26	44.15
	14 days	3.51	2.64	6.79	-0.92	45.52
Transformer-Time2Vector	1 day	2.27	1.62	3.98	0.33	44.58
	7 days	2.93	2.09	5.32	-0.09	46.25
	14 days	3.58	2.61	6.73	-0.82	45.75
MST-Net	**1 day**	**1.65**	**1.29**	**3.04**	**0.60**	**44.64**
	7 days	**2.60**	**1.81**	**4.57**	**0.07**	**45.42**
	14 days	**3.33**	**2.36**	**6.01**	**0.02**	**46.19**

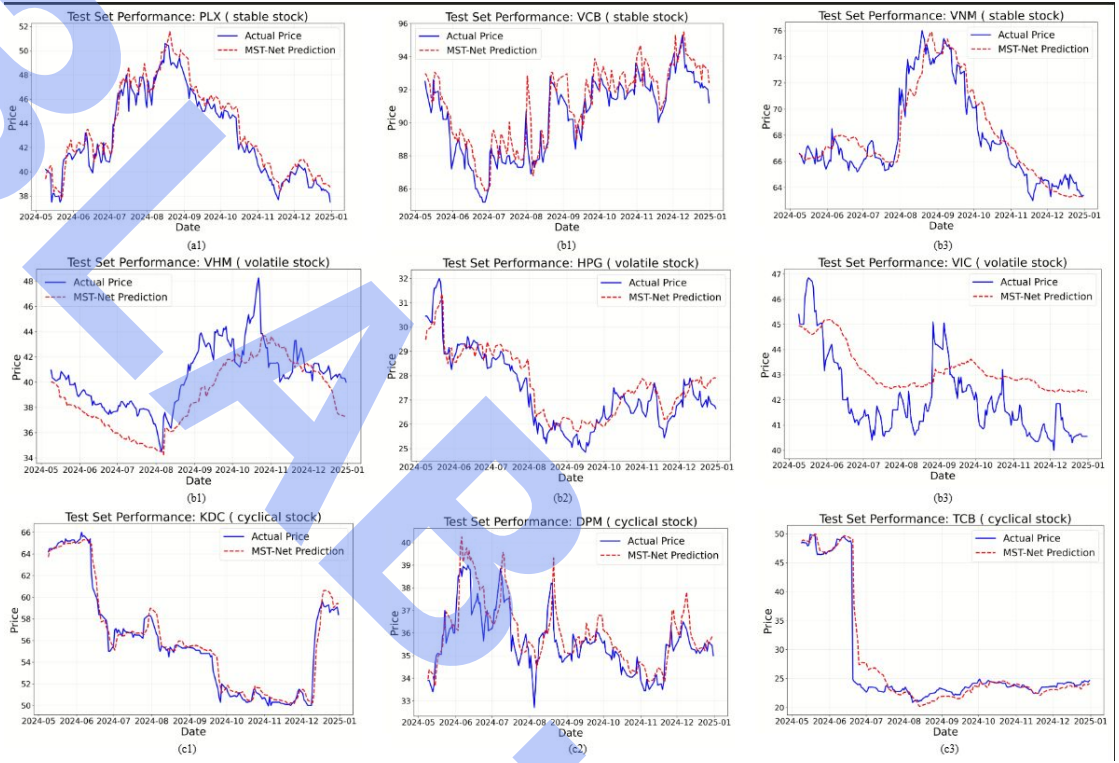
S3LAB

Stock Data Analysis

Analyze the Data - Predictive

Prediction performance across different stock behaviors

- Stable stocks: High accuracy with minimal deviation
- Volatile stocks: Captures major price movements despite volatility
- Cyclical stocks: Successfully tracks pattern changes



SPECIAL

Stock Data Analysis

Analyze the Data - Prescriptive

- Predictive models such as MST-Net can forecast future prices but cannot make trading decisions.
- Financial markets require prescriptive intelligence — systems that decide what action to take under uncertainty.

=> **Our approach**: Deep Reinforcement Learning

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Stock Data Analysis

Analyze the Data - Prescriptive

Reinforcement Learning Agent

learns optimal trading strategies through continuous interaction with the market:

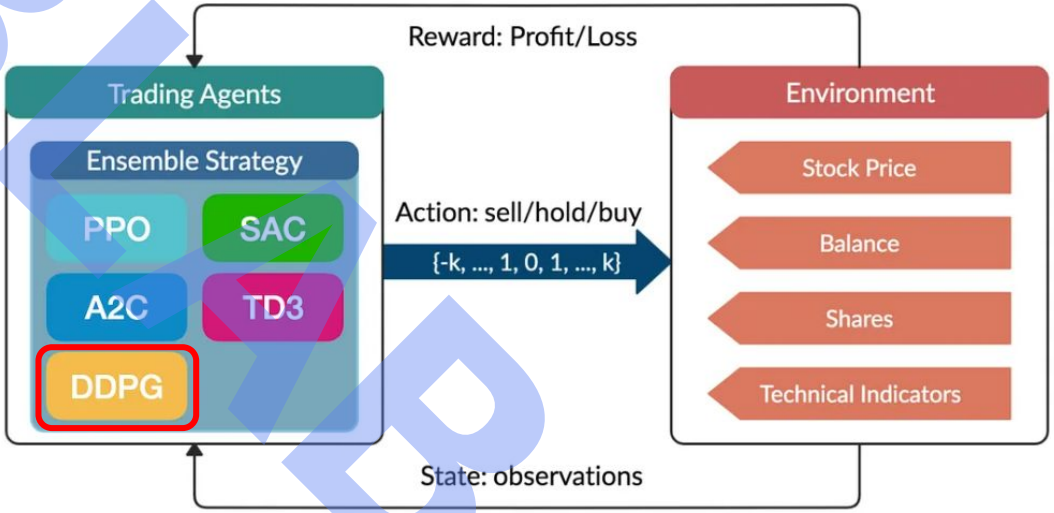
- Observes current and predicted market states.
- Executes actions (**Buy +1**, Hold **0**, or Sell **-1**) to maximize long-term portfolio return.
- Receives rewards based on profit, risk, and transaction cost.
- Continuously improves its policy through feedback from the environment.

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Stock Data Analysis

Analyze the Data - Prescriptive

1. Normalized prices and returns (4 features)
2. Technical indicators: MA5, MA10, MA20, RSI, MACD components (7 features)
3. Portfolio status: Cash ratio, Stock ratio, position (3 features)
4. Market context: VN-Index return, volatility regime (2 features)



Deep Reinforcement Learning for Automated Stock Trading

Theory reference: Deep Reinforcement Learning for Automated Stock Trading

Stock Data Analysis

Analyze the Data - Prescriptive

Algorithm ("agent")	Description	Policy Type	Action Space	Sample Efficiency	Exploration Strategy	Stability	Complexity	Common Applications
PPO	Proximal Policy Optimization - balances simplicity and performance	On-policy	Discrete, Continuous	Moderate	Clipped Surrogate Objective	High	Medium	Robotics, Games, Continuous control
A2C	Advantage Actor-Critic - synchronous version of A3C	On-policy	Discrete, Continuous	Moderate	Entropy Regularization	Moderate	Low	Robotics, Games, Continuous control
DDPG	Deep Deterministic Policy Gradient - off-policy actor-critic for continuous action spaces	Off-policy	Continuous	High	Ornstein-Uhlenbeck Noise	Moderate	High	Robotics, Continuous control tasks
SAC	Soft Actor-Critic - maximizes entropy to improve exploration	Off-policy	Continuous	High	Entropy Regularization	High	High	Robotics, Continuous control tasks
TD3	Twin Delayed Deep Deterministic Policy Gradient - addresses overestimation bias in DDPG	Off-policy	Continuous	High	Target Policy Smoothing	High	High	Robotics, Continuous control tasks

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Limitations of Traditional DRL in Stock Trading

- Focuses on short-term profit, ignoring risk and market dynamics.
- Applies one reward function for all stocks → lacks adaptability.
- Unstable learning under non-stationary and high-cost markets.
- Limited transparency in trading decisions.

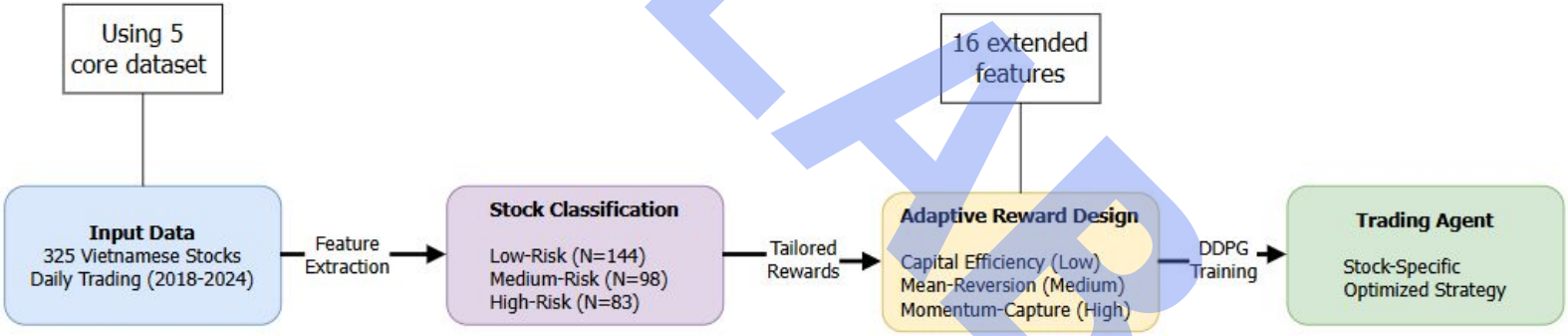
→ **These issues motivate our adaptive framework — MP-DRE.**

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MP-DRE — a multi-phase DRL framework designed for adaptive and risk-aware stock trading.



Key Insight: Different stock behaviors require different reward structures

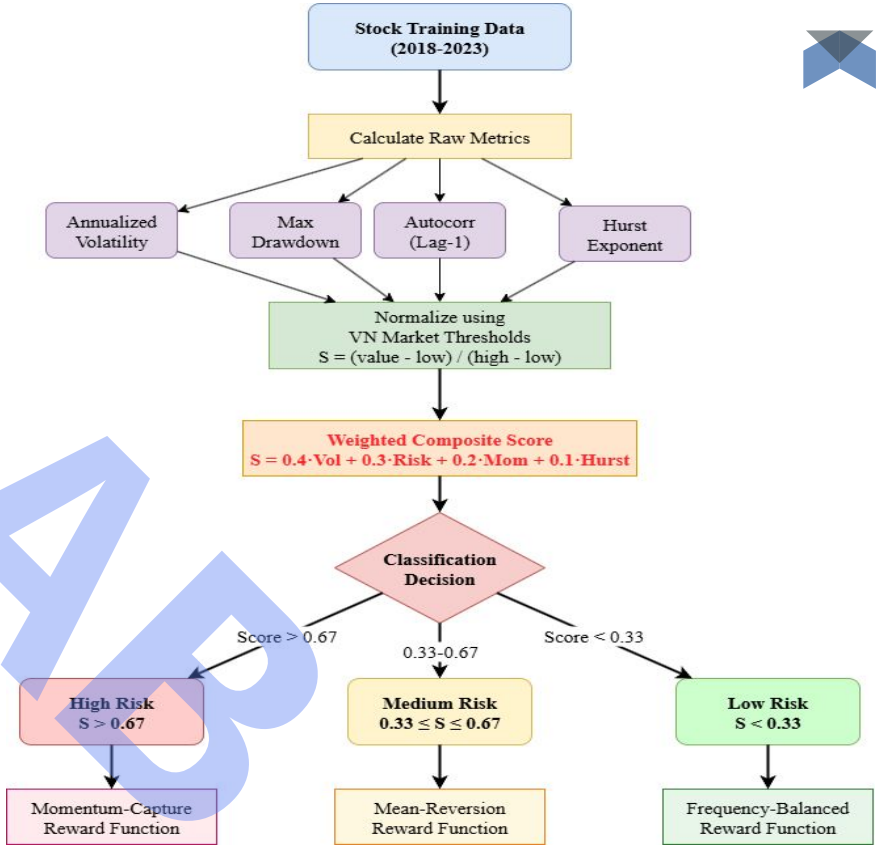
Our framework integrates classification and adaptive reward before training the DRL agent. This allows each stock group to learn differently based on its risk profile.

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A **risk-based stock classification module** to segment stocks into **Low-, Medium-, and High-Risk** categories for adaptive reward learning.



Stock classification pipeline for adaptive reward mapping

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Risk-specific reward functions based on stock classification results.

Table 2: Stock-Type-Specific Reward Components

Risk Class	Weight	Component Description
Low-Risk	0.40	Profit: Portfolio value change Δ_{asset}
	0.25	Strategy consistency: Penalty for position changes
	0.15	Risk control: Drawdown penalty (threshold 5%)
	0.20	Efficiency: Transaction cost penalty (0.3%)
Medium-Risk	0.30	Profit: Portfolio value change Δ_{asset}
	0.35	Mean-reversion: Reward contrarian positioning vs MA_{20}
	0.15	Regime penalty: Penalize holding during extreme moves ($> 3\%$)
	0.20	Efficiency: Transaction cost penalty
High-Risk	0.60	Momentum: Reward directional continuation
	0.25	Volatility-adjusted: Amplify gains during high-vol regimes
	0.15	Efficiency: Transaction cost penalty

Note: Weights reflect the relative importance of each component. Low-Risk emphasizes capital efficiency, Medium-Risk exploits mean-reversion, and High-Risk captures momentum trends.

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Our results

Table 3: Performance by Risk Category (Averaged Over Test Stocks, 2024)

Risk Class	Method	Profit (%)	Sharpe	Sortino	Max DD (%)	Good Days
Low (N=3)	DDPG (Ours)	33.36	1.24	1.87	14.77	18
	Moving Average	15.18	0.78	1.29	19.67	16
	Signal Rolling	21.73	1.78	2.26	9.12	24
Med (N=3)	DDPG (Ours)	**235.63**	**2.38**	**2.84**	8.39	51
	Moving Average	111.07	0.66	1.45	41.84	22
	Signal Rolling	28.68	1.67	1.73	8.50	26
High (N=3)	DDPG (Ours)	71.48	1.44	1.87	25.06	79
	Moving Average	64.63	1.99	2.35	37.13	24
	Signal Rolling	20.08	1.18	1.06	8.62	23

Note: Results represent average performance across 3 representative stocks per risk category.

DDPG (Ours) uses adaptive rewards tailored to each risk class. Good Days counts profitable trading sessions.

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Q & A



Thank you for listening

"Coming together is a beginning;
Keeping together is progress;
Working together is success."
- HENRY FORD

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